Thomas Fermi Equations

We have two Thomas Fermi equations as

$$\overline{\alpha}_I \frac{d^2\overline{f_I(x)}}{dx^2} = -\frac{6\eta}{5(5\eta-1)} \sum_J g_J \frac{(f_J(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{1}$$

$$\alpha_I \frac{d^2 f_I(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)} \sum_I \overline{g}_I \frac{\left(\overline{f_I(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)} \left[\frac{(N_I g_I - 1)}{N_I} \frac{(f_I(x))^{\frac{3}{2}}}{\sqrt{x}} + \sum_{I \neq J} g_J \frac{(f_J(x))^{\frac{3}{2}}}{\sqrt{x}} \right] \tag{2}$$

With the work done by Giri *et.al* we know that the distribution of quarks inside a pocket of pentaquark is divided into three different regions.

Above two equations can be expanded with three different functions namely $f_1(x)$ for light quarks (uud), $f_2(x)$ for heavy quark c & $\overline{f_2(x)}$ for heavy anti-quark $\bar{c}$ as

$$\overline{\alpha}_2 \frac{d^2\overline{f_2(x)}}{dx^2} = -\frac{6\eta}{5(5\eta-1)} \left\{ g_1 \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2 \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \right\} \tag{3}$$

$$\alpha_1 \frac{d^2 f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)} \left[\frac{(N_1 g_1 - 1)}{N_1} \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2 \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \right] \tag{4}$$

$$\alpha_2 \frac{d^2 f_2(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)} \overline{g_2} \frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)} \left[\frac{(N_2 g_2 - 1)}{N_2} \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_1 \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \right] \tag{5}$$

We now have to be clear in the three different regions which functions are influential.

Case I

$0 < x \leq \overline{x_2}$, all three functions are present but anti-charm quark is dominant

$\overline{x_2} < x \leq x_2$, only charm quark and light quarks are present and charm quark is dominant

$x_2 < x \leq x_1$, only light quark is present.

Therefore, our study from now on involves treating equations (3), (4), (5) in these three different regions.

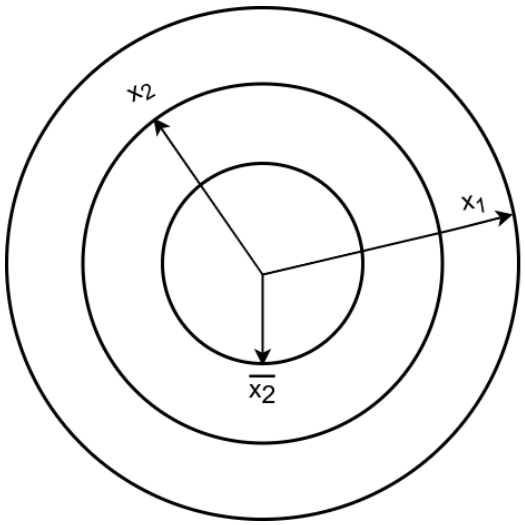


Figure 1: Distribution of quarks for the case of $\eta = 1$

- **In region** $x_2 < x \leq x_1$

In this region, $f_2(x) = 0$ & $\overline{f_2(x)} = 0$.

Equation 3 gives

$$0 = -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right\} \tag{6}$$

Equation 4 gives

$$\alpha_1\frac{d^2f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1g_1-1)}{N_1}\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right] \tag{7}$$

Equation 5 gives

$$0 = -\frac{9\eta}{10(5\eta-1)}\left[g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right] \tag{8}$$

Equations (123), (125) results $f_1(x) = 0$, which leads to meaningless result. So we drop these two equations and consider only equation (124).

$$\alpha_1\frac{d^2f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1g_1-1)}{N_1}\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

We have, $\alpha_1 = \frac{m^1}{m^1} = 1$. So that,

$$\frac{d^2f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1g_1-1)}{N_1}\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

$$\frac{d^2f_1(x)}{dx^2} = Q_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{9}$$

$$\text{where, } Q_1 = \frac{-9\eta}{10(5\eta-1)}\frac{(N_1g_1-1)}{N_1} \tag{10}$$

- **In region** $0 < x \leq \overline{x_2}$

In this region all three functions are present but we use linear relationship of other functions with $\overline{f_2(x)}$ as

$$\begin{aligned} f_1(x) &= \beta_{1\overline{2}}\overline{f_2(x)} \\ f_2(x) &= \beta_{2\overline{2}}\overline{f_2(x)} \end{aligned} \tag{11}$$

where, $\beta_{1\overline{2}}$ & $\beta_{2\overline{2}}$ are proportionality constants.

Equation 3 gives

$$\overline{\alpha}_2\frac{d^2\overline{f_2(x)}}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right\}$$

$$\overline{\alpha}_2\frac{d^2\overline{f_2(x)}}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(\beta_{1\overline{2}}\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(\beta_{2\overline{2}}\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\right\}$$

$$\frac{d^2\overline{f_2(x)}}{dx^2} = -\frac{6\eta}{5(5\eta-1)\overline{\alpha}_2}\left\{g_1(\beta_{1\overline{2}})^{\frac{3}{2}} + g_2(\beta_{2\overline{2}})^{\frac{3}{2}}\right\}\frac{\overline{f_2(x)}^{\frac{3}{2}}}{\sqrt{x}} \tag{12}$$

Equation 4 gives

$$\alpha_1\frac{d^2f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1g_1-1)}{N_1}\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

We have, $\alpha_1 = \frac{m^1}{m^1} = 1$. So that,

$$\beta_{1\overline{2}}\frac{d^2\overline{f_2(x)}}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1g_1-1)(\beta_{1\overline{2}})^{\frac{3}{2}}}{N_1}\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} + g_2(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}}\right]$$

$$\frac{d^2\overline{f_2(x)}}{dx^2} = -\frac{9\eta}{10(5\eta-1)\beta_{1\overline{2}}}\left[\frac{(N_1g_1-1)(\beta_{1\overline{2}})^{\frac{3}{2}}}{N_1} + g_2(\beta_{2\overline{2}})^{\frac{3}{2}}\right]\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} \tag{13}$$

Equation 5 gives

$$\alpha_2\frac{d^2f_2(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\overline{g_2}\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)}\left[\frac{(N_2g_2-1)}{N_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

$$\alpha_2\beta_{2\overline{2}}\frac{d^2\overline{f_2(x)}}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\overline{g_2}\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)}\left[\frac{(N_2g_2-1)(\beta_{2\overline{2}})^{\frac{3}{2}}}{N_2}\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} + g_1(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}}\right]$$

$$\frac{d^2\overline{f_2(x)}}{dx^2} = \left[-\frac{6\eta}{5(5\eta-1)\alpha_2\beta_{2\overline{2}}}\overline{g_2} - \frac{9\eta}{10(5\eta-1)\alpha_2\beta_{2\overline{2}}}\left\{\frac{(N_2g_2-1)(\beta_{2\overline{2}})^{\frac{3}{2}}}{N_2} + g_1(\beta_{1\overline{2}})^{\frac{3}{2}}\right\}\right]\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} \tag{14}$$

In a single form eq^{n}s (12), (13), (14) can be written as

$$\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{Q_2}\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} \tag{15}$$

where,

$$\begin{aligned}
&= -\frac{6\eta}{5(5\eta-1)\overline{\alpha}_2}\left\{g_1(\beta_{1\overline{2}})^{\frac{3}{2}} + g_2(\beta_{2\overline{2}})^{\frac{3}{2}}\right\} \\
\overline{Q_2} &= -\frac{9\eta}{10(5\eta-1)\beta_{1\overline{2}}}\left[\frac{(N_1g_1-1)(\beta_{1\overline{2}})^{\frac{3}{2}}}{N_1} + g_2(\beta_{2\overline{2}})^{\frac{3}{2}}\right] \\
&= -\frac{6\eta}{5(5\eta-1)\alpha_2\beta_{2\overline{2}}}\overline{g_2} - \frac{9\eta}{10(5\eta-1)\alpha_2\beta_{2\overline{2}}}\left\{\frac{(N_2g_2-1)(\beta_{2\overline{2}})^{\frac{3}{2}}}{N_2} + g_1(\beta_{1\overline{2}})^{\frac{3}{2}}\right\}
\end{aligned} \tag{16}$$

- **In region** $\overline{x_2} < x \leq x_2$

In this region, two functions $f_2(x)$ & $f_1(x)$ are present and we use linear relationship of $f_1(x)$ with $f_2(x)$ as

$$f_1(x) = \gamma_{12}f_2(x) \tag{17}$$

Equation 3 gives

$$\begin{aligned}
\overline{\alpha}_2\frac{d^2\overline{f_2(x)}}{dx^2} &= -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right\} \\
0 &= -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right\} \\
0 &= \left\{g_1(\gamma_{12})^{\frac{3}{2}} + g_2\right\}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \\
\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} &= 0
\end{aligned} \tag{18}$$

This leads to meaningless result. So we drop this equation.

Equation 4 gives

$$\alpha_1\frac{d^2f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1g_1-1)}{N_1}\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

We have, $\alpha_1 = \frac{m^1}{m^1} = 1$. So that,

$$\gamma_{12}\frac{d^2f_2(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1g_1-1)(\gamma_{12})^{\frac{3}{2}}}{N_1}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

$$\frac{d^2f_2(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)\gamma_{12}}\left\{\frac{(N_1g_1-1)(\gamma_{12})^{\frac{3}{2}}}{N_1} + g_2\right\}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{19}$$

Equation 5 gives

$$\alpha_2\frac{d^2f_2(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\overline{g_2}\frac{\left(\overline{f_2(x)}\right)^{\frac{3}{2}}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)}\left[\frac{(N_2g_2-1)}{N_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

$$\alpha_2\frac{d^2f_2(x)}{dx^2} = 0 - \frac{9\eta}{10(5\eta-1)}\left[\frac{(N_2g_2-1)}{N_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_1(\gamma_{12})^{\frac{3}{2}}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

$$\frac{d^2f_2(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)\alpha_2}\left\{\frac{(N_2g_2-1)}{N_2} + g_1(\gamma_{12})^{\frac{3}{2}}\right\}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{20}$$

In a single form eq^{n}s (19), (20) can be written as

$$\frac{d^2f_2(x)}{dx^2} = Q_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{21}$$

where,

$$\begin{aligned} Q_2 &= -\frac{9\eta}{10(5\eta-1)\gamma_{12}}\left\{\frac{(N_1g_1-1)(\gamma_{12})^{\frac{3}{2}}}{N_1} + g_2\right\} \\ &= -\frac{9\eta}{10(5\eta-1)\alpha_2}\left\{\frac{(N_2g_2-1)}{N_2} + g_1(\gamma_{12})^{\frac{3}{2}}\right\} \end{aligned} \tag{22}$$

Therefore, eq^{n}s (126), (15), (21) are the required consistency conditions with the values of Q_1, $\overline{Q_2}$ & Q_2 in eq^{n}s (127), (16) & (22) repectively.

Calculation of Kinetic Energy

$$T = \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3 m^I (g^I)^{\frac{2}{3}}} \left(n^I(r)\right)^{\frac{5}{3}} + \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3 \overline{m}^I (\overline{g^I})^{\frac{2}{3}}} \left(\overline{n^I(r)}\right)^{\frac{5}{3}} \tag{23}$$

We have,

$$n^I(r) = \left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{3}{2}} \left(\frac{f^I(r)}{r}\right)^{\frac{3}{2}} \frac{g^I}{6\pi^2} \tag{24}$$

$$\overline{n^I(r)} = \left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{3}{2}} \left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{3}{2}} \frac{\overline{g}^I}{6\pi^2} \tag{25}$$

So that,

$$\begin{aligned} T = &\sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3 m^I (g^I)^{\frac{2}{3}}} \left[\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{3}{2}} \left(\frac{f^I(r)}{r}\right)^{\frac{3}{2}} \frac{g^I}{6\pi^2}\right]^{\frac{5}{3}} \\ &+ \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3 \overline{m}^I (\overline{g^I})^{\frac{2}{3}}} \left[\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{3}{2}} \left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{3}{2}} \frac{\overline{g}^I}{6\pi^2}\right]^{\frac{5}{3}} \\ &\implies T = T_1 + T_2 \end{aligned} \tag{26}$$

Taking T_1,

$$T_1 = \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3 m^I (g^I)^{\frac{2}{3}}} \left[\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{3}{2}} \left(\frac{f^I(r)}{r}\right)^{\frac{3}{2}} \frac{g^I}{6\pi^2}\right]^{\frac{5}{3}}$$

$$= \sum_I \int^{r_{max}} d^3r \frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3 m^I (g^I)^{\frac{2}{3}}} \left(\frac{g^I}{6\pi^2}\right)^{\frac{5}{3}} \left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{3}{2}\times\frac{5}{3}} \left(\frac{f^I(r)}{r}\right)^{\frac{3}{2}\times\frac{5}{3}}$$

$$= \sum_I \int^{r_{max}} d^3r \frac{3\hbar^{3\times\frac{5}{3}} (g^I)^{\frac{5}{3}-\frac{2}{3}}}{20\pi^2\hbar^3 m^I} \left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{5}{2}} \left(\frac{f^I(r)}{r}\right)^{\frac{5}{2}}$$

$$= \frac{3\hbar^2}{20\pi^2 m_1} \left\{\frac{2\times(\frac{4}{3})\alpha_s}{a}\right\}^{\frac{5}{2}} 4\pi \sum_I \int^{r_{max}} dr r^2 \alpha_I g^I \left(\frac{f_I(r)}{r}\right)^{\frac{5}{2}}$$

where, $\alpha_I = \frac{m_1}{m_I}$

$$= constant. \sum_I \int^{r_{max}} dr r^2 \alpha_I g^I \left(\frac{f_I(r)}{r}\right)^{\frac{5}{2}}$$

Put $r = Rx$, $dr = Rdx$

$$= constant. \sum_I \int^{x_{max}} Rdx R^2 x^2 \alpha_I g^I \left(\frac{f^I(x)}{Rx}\right)^{\frac{5}{2}}$$

$$= constant. \sum_I \int^{x_{max}} dx R^{3-\frac{5}{2}} \alpha_I g^I \frac{(f^I(x))^{\frac{5}{2}}}{x^{\frac{5}{2}-2}}$$

$$= constant\sqrt{R}. \sum_I \int^{x_{max}} dx \alpha_I g^I \frac{(f^I(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$T_1 = C \sum_I \int^{x_{max}} dx \alpha_I g^I \frac{(f^I(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$\frac{T_1}{C} = \sum_I \int^{x_{max}} dx \alpha_I g^I \frac{(f^I(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \alpha_1 g_1 \int_0^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_2 g_2 \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \alpha_1 g_1 \int_0^{\overline{x_2}} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_{\overline{x_2}}^{x_2} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_2 g_2 \int_0^{\overline{x_2}} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_2 g_2 \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \alpha_1 g_1 (\beta_{1\overline{2}})^{\frac{5}{2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 (\gamma_{12})^{\frac{5}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_2 g_2 (\beta_{2\overline{2}})^{\frac{5}{2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$
$$+ \alpha_2 g_2 \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$\longrightarrow \frac{T_1}{C} = \left[\alpha_1 g_1 (\beta_{1\overline{2}})^{\frac{5}{2}} + \alpha_2 g_2 (\beta_{2\overline{2}})^{\frac{5}{2}}\right] \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \left[\alpha_I g_1 (\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\right] \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} \tag{27}$$

Taking T_2,

$$T_2 = \sum_I \int^{r_{max}} d^3 r \frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3 \overline{m}^I (\overline{g^I})^{\frac{2}{3}}} \left[\left\{\frac{2 \times (4/3)\alpha_s}{a}\right\}^{\frac{3}{2}} \left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{3}{2}} \frac{\overline{g}^I}{6\pi^2}\right]^{\frac{5}{3}}$$

$$= \sum_I \int^{r_{max}} d^3 r \frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3 \overline{m}^I (\overline{g^I})^{\frac{2}{3}}} \left(\frac{\overline{g^I}}{6\pi^2}\right)^{\frac{5}{3}} \left\{\frac{2 \times (4/3)\alpha_s}{a}\right\}^{\frac{3}{2}\times\frac{5}{3}} \left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{3}{2}\times\frac{5}{3}}$$

$$= \sum_I \int^{r_{max}} d^3 r \frac{3\hbar^2 \overline{g^I}}{20\pi^2 \overline{m^I}} \left\{\frac{2 \times (4/3)\alpha_s}{a}\right\}^{\frac{5}{2}} \left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{5}{2}}$$

$$= \frac{3\hbar^2}{20\pi^2} \left\{\frac{2 \times (4/3)\alpha_s}{a}\right\}^{\frac{5}{2}} \sum_I \int^{r_{max}} d^3 r \frac{\overline{g^I}}{\overline{m^I}} \left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{5}{2}}$$

$$= \frac{3\hbar^2}{20\pi^2 m_1} \left\{\frac{2 \times (4/3)\alpha_s}{a}\right\}^{\frac{5}{2}} 4\pi \sum_I \int^{r_{max}} dr r^2 \overline{\alpha_I} \overline{g^I} \left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{5}{2}}$$

where, $\overline{\alpha_I} = \frac{m_1}{\overline{m_I}}$

$$= constant. \sum_I \int^{r_{max}} dr r^2 \overline{\alpha_I} \overline{g^I} \left(\frac{\overline{f^I(r)}}{r}\right)^{\frac{5}{2}}$$

Put $r = Rx$, $dr = Rdx$

$$= constant. \sum_I \int^{x_{max}} Rdx R^2 x^2 \overline{\alpha_I g_I} \left(\frac{\overline{f^I(x)}}{Rx}\right)^{\frac{5}{2}}$$

$$= constant. \sum_I \int^{x_{max}} dx \sqrt{R} \overline{\alpha_I g_I} \frac{(\overline{f^I(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$= constant \sqrt{R}. \sum_I \int^{x_{max}} dx \overline{\alpha_I g_I} \frac{(\overline{f^I(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$T_2 = C \sum_I \int^{x_{max}} dx \overline{\alpha_I g_I} \frac{(\overline{f^I(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$\frac{T_2}{C} = \sum_I \int^{x_{max}} dx \overline{\alpha_I g_I} \frac{(\overline{f^I(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$\longrightarrow \frac{T_2}{C} = \overline{\alpha_2 g_2} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} \tag{28}$$

Adding eq^{n}s (27) & (28),

$$\frac{T_1}{C} + \frac{T_2}{C} = \left[\alpha_1 g_1 (\beta_{1\overline{2}})^{\frac{5}{2}} + \alpha_2 g_2 (\beta_{2\overline{2}})^{\frac{5}{2}}\right] \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \left[\alpha_1 g_1 (\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\right] \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$
$$+ \alpha_1 g_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \overline{\alpha_2 g_2} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \left[\alpha_1 g_1(\beta_{1\overline{2}})^{\frac{5}{2}} + \alpha_2 g_2(\beta_{2\overline{2}})^{\frac{5}{2}} + \overline{\alpha_2 g_2}\right] \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \left[\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\right] \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}$$

From appendix,

$$\int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \tag{29}$$

$$\begin{aligned}\int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} = {} & \frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5}{7Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}\left[\frac{N_2Q_2}{3\eta} + \frac{\beta_{1\overline{2}}}{\gamma_{12}}\frac{\overline{N_2Q_2}}{3\eta} - \right. \\ & \left.(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\right] + \frac{5}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\frac{\overline{N_2Q_2}}{3\eta}\end{aligned} \tag{30}$$

$$\begin{aligned}\int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} = {} & \frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} - \frac{4}{7}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5}{7Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}\left[\frac{N_1Q_1}{3\eta} + \gamma_{12}\frac{N_2Q_2}{3\eta} - (\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}\right. \\ & \left. + \frac{\overline{N_2}Q_1}{3\eta}\left\{(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - (\beta_{1\overline{2}})^{\frac{3}{2}}\right\} - (\gamma_{12})(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta} + (\gamma_{12})\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta}\right] \\ & + \frac{5}{7Q_1}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}\left[\frac{N_2Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\right]\end{aligned} \tag{31}$$

So that,

$$\begin{aligned}\frac{T_1}{C} + \frac{T_2}{C} = {} & \left[\alpha_1 g_1(\beta_{1\overline{2}})^{\frac{5}{2}} + \alpha_2 g_2(\beta_{2\overline{2}})^{\frac{5}{2}} + \overline{\alpha_2 g_2}\right]\left[\frac{4}{7}\sqrt{x}\,(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] + \left[\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\right]\left[\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}\right. \\ & - \frac{4}{7}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5}{7Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}\left[\frac{N_2Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\right] + \frac{5}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2 \\ & \left.\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\frac{\overline{N_2Q_2}}{3\eta}\right] + \alpha_1 g_1\left[\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} - \frac{4}{7}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5}{7Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}\left[\frac{N_1Q_1}{3\eta} + \gamma_{12}\frac{N_2Q_2}{3\eta}\right.\right. \\ & - (\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta} + \frac{\overline{N_2}Q_1}{3\eta}\left\{(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - (\beta_{1\overline{2}})^{\frac{3}{2}}\right\} - (\gamma_{12})(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta} + (\gamma_{12})\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta}\Big] \\ & \left.\left. + \frac{5}{7Q_1}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}\left[\frac{N_2Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\right]\right]\end{aligned}$$

$$\begin{aligned}= {} & \frac{4}{7}\left[\alpha_1 g_1(\beta_{1\overline{2}})^{\frac{5}{2}} + \alpha_2 g_2(\beta_{2\overline{2}})^{\frac{5}{2}} + \overline{\alpha_2 g_2}\right]\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta}\left[\alpha_1 g_1(\beta_{1\overline{2}})^{\frac{5}{2}} + \alpha_2 g_2(\beta_{2\overline{2}})^{\frac{5}{2}} + \overline{\alpha_2 g_2}\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\ & + \frac{4}{7}\left[\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\right]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\left[\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\right]\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5}{7Q_2}\left[\frac{N_2Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta}\right. \\ & \left. - (\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\right]\left[\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{\overline{N_2Q_2}}{3\eta}\left[\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\ & + \frac{4}{7}\alpha_1 g_1\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} - \frac{4}{7}\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5\alpha_1 g_1}{7Q_1}\left\{\frac{N_1Q_1}{3\eta} + \gamma_{12}\frac{N_2Q_2}{3\eta} - (\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}\right. \\ & + (\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta} - (\gamma_{12})(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta} + (\gamma_{12})\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta}\Big\}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5}{7Q_1}(\gamma_{12})^2\alpha_1 g_1\Big\{\frac{N_2Q_2}{3\eta} \\ & \left. + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\Big\}\frac{df_2(x)}{dx}\Big|_{x_2}\right.\end{aligned}$$

$$\frac{(T_1+T_2)}{C} = \frac{4}{7}\big[\alpha_1 g_1(\beta_{1\bar{2}})^{\frac{5}{2}} + \alpha_2 g_2(\beta_{2\bar{2}})^{\frac{5}{2}} + \overline{\alpha_2 g_2} - \alpha_1 g_1(\gamma_{12})^{\frac{5}{2}}\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)^{\frac{5}{2}} - \alpha_2 g_2\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\big]\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7}\big[\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}}$$
$$+ \alpha_2 g_2 - \alpha_1 g_1(\gamma_{12})^{\frac{5}{2}}\big]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7}\alpha_1 g_1\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \frac{5\overline{N_2}}{21\eta}\Bigg[\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)^2\frac{\overline{Q_2}}{Q_2}\big\{\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\big\}$$
$$- \alpha_1 g_1(\beta_{1\bar{2}})^{\frac{5}{2}} - \alpha_2 g_2(\beta_{2\bar{2}})^{\frac{5}{2}} - \overline{\alpha_2 g_2}\Bigg]\frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{5}{21\eta}\Bigg[\frac{(\gamma_{12})^2\alpha_1 g_1}{Q_1}\Bigg\{N_2Q_2 + \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)\overline{N_2Q_2} - (\beta_{2\bar{2}})^{\frac{3}{2}}\overline{N_2}Q_2\Bigg\}$$
$$- \big\{\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\big\}\Bigg\{N_2 + \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{Q_2} - (\beta_{2\bar{2}})^{\frac{3}{2}}\overline{N_2}\Bigg\}\Bigg]\frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{5\alpha_1 g_1}{21\eta}\Bigg[N_1 + \gamma_{12}\frac{N_2Q_2}{Q_1}$$
$$+ (\gamma_{12})^{\frac{3}{2}}\big\{(\beta_{2\bar{2}})^{\frac{3}{2}}\overline{N_2} - N_2\big\} - (\beta_{1\bar{2}})^{\frac{3}{2}}\overline{N_2} - (\gamma_{12})(\beta_{2\bar{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{Q_1} + (\gamma_{12})\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{Q_1}\Bigg]\frac{df_1(x)}{dx}\Bigg|_{x_1}$$

Therefore, the final expression of kinetic energy is

$$\frac{(K.E)}{C} = \frac{4}{7}\Bigg[\alpha_1 g_1\big\{(\beta_{1\bar{2}})^{\frac{5}{2}} - (\gamma_{12})^{\frac{5}{2}}\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\big\} + \overline{\alpha_2 g_2}\Bigg]\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7}\alpha_2 g_2\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7}\alpha_1 g_1\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}$$
$$+ \frac{5\overline{N_2}}{21\eta}\Bigg[\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)^2\frac{\overline{Q_2}}{Q_2}\big\{\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\big\} - \alpha_1 g_1(\beta_{1\bar{2}})^{\frac{5}{2}} - \alpha_2 g_2(\beta_{2\bar{2}})^{\frac{5}{2}} - \overline{\alpha_2 g_2}\Bigg]\frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{5}{21\eta}\Bigg[\frac{(\gamma_{12})^2\alpha_1 g_1}{Q_1}$$
$$\Bigg\{N_2Q_2 + \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)\overline{N_2Q_2} - (\beta_{2\bar{2}})^{\frac{3}{2}}\overline{N_2}Q_2\Bigg\} - \Bigg\{\alpha_1 g_1(\gamma_{12})^{\frac{5}{2}} + \alpha_2 g_2\Bigg\}\Bigg\{N_2 + \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{Q_2} - (\beta_{2\bar{2}})^{\frac{3}{2}}\overline{N_2}\Bigg\}\Bigg]\frac{df_2(x)}{dx}\Bigg|_{x_2}$$
$$- \frac{5\alpha_1 g_1}{21\eta}\Bigg[N_1 + \gamma_{12}\frac{N_2Q_2}{Q_1} + (\gamma_{12})^{\frac{3}{2}}\Bigg\{(\beta_{2\bar{2}})^{\frac{3}{2}}\overline{N_2} - N_2\Bigg\} - (\beta_{1\bar{2}})^{\frac{3}{2}}\overline{N_2} - (\gamma_{12})(\beta_{2\bar{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{Q_1}$$
$$+ (\gamma_{12})\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{Q_1}\Bigg]\frac{df_1(x)}{dx}\Bigg|_{x_1} \tag{32}$$

Where,

$$C = \frac{12\hbar^2\sqrt{R}}{20\pi m_1}\left\{\frac{2\times(4/3)\alpha_s}{a}\right\}^{\frac{5}{2}}$$

Appendix for Kinetic Energy

1. Noramalization condition gives

$$\int_0^{\overline{x_{max}}} dx\sqrt{x}(\overline{f_I(x)})^{\frac{3}{2}} = \frac{\overline{N_I}}{3\eta} \tag{33}$$
$$\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta}$$

2. The Thomas - Fermi equation gives

$$\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{Q_2}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$
$$x\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{Q_2}\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$
$$\int_0^{\overline{x_2}} x\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{Q_2}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$
$$\int_0^{\overline{x_2}} x\frac{d^2\overline{f_2(x)}}{dx^2} = \frac{\overline{Q_2N_2}}{3\eta}$$
$$\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}} = \frac{\overline{N_2Q_2}}{3\eta} \tag{34}$$

3. Normalization condition gives

$$\int_0^{x_{max}} dx\sqrt{x}(f_I(x))^{\frac{3}{2}} = \frac{N_I}{3\eta}$$

$$\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} = \frac{N_2}{3\eta}$$

$$\int_0^{\overline{x_2}} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} = \frac{N_2}{3\eta}$$

$$(\beta_{2\overline{2}})^{\frac{3}{2}} \int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} = \frac{N_2}{3\eta}$$

$$(\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta} + \int_{\overline{x_2}}^{x_2} dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2} = \frac{N_2}{3\eta}$$

$$\frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx x \frac{d^2 f_2(x)}{dx^2} = \frac{N_2}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}}^{x_2} = \frac{N_2}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}} = \frac{N_2 Q_2}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta}$$

$$\left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \beta_{2\overline{2}} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}} = \frac{N_2 Q_2}{3\eta} - \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta}$$

$$\left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{\overline{N_2 Q_2}}{3\eta} = \frac{N_2 Q_2}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta}$$

$$\left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} = \frac{N_2 Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{\overline{N_2 Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta} \tag{35}$$

4. Normalization condition gives

$$\int_0^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \frac{N_1}{3\eta}$$

$$\int_{)}^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} + \int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \frac{N_1}{3\eta}$$

$$(\beta_{1\overline{2}})^{\frac{3}{2}} \int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} + (\gamma_{12})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} + \int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \frac{N_1}{3\eta}$$

$$(\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta} + (\gamma_{12})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2} + \int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \frac{N_1}{3\eta}$$

$$\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \int_{\overline{x_2}}^{x_2} dx x \frac{d^2 f_2(x)}{dx^2} + \int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \frac{N_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}}^{x_2} + \int_{x_2}^{x_1} dx \frac{x}{Q_1} \frac{d^2 f_1(x)}{dx^2} = \frac{N_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left\{ \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}} \right\} + \frac{1}{Q_1} \int_{x_2}^{x_1} dx x \frac{d^2 f_1(x)}{dx^2} = \frac{N_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left\{ \frac{N_2 Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{\overline{N_2 Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta} - \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \left(x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right)_{\overline{x_2}} \right\} + \frac{1}{Q_1} \int_{x_2}^{x_1} dx x \frac{d^2 f_1(x)}{dx^2} = \frac{N_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left\{ \frac{N_2 Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{\overline{N_2 Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta} - \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{\overline{N_2 Q_2}}{3\eta} \right\} + \frac{1}{Q_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_2}^{x_1} = \frac{N_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left\{ \frac{N_2 Q_2}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta} \right\} + \frac{1}{Q_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_2}^{x_1} = \frac{N_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\frac{1}{Q_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_2}^{x_1} = \frac{N_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \cdot \frac{N_2 Q_2}{3\eta} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \cdot (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta}$$

$$\frac{1}{Q_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_2}^{x_1} = \frac{N_1}{3\eta} - (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta} - (\gamma_{12})^{\frac{3}{2}} \frac{N_2}{3\eta} + (\gamma_{12})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta}$$

$$\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_1}-\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_2}=\frac{N_1Q_1}{3\eta}-(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}-(\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}$$

$$\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_1}-\gamma_{12}\left[x\frac{df_2(x)}{dx}-f_2(x)\right]_{x_2}=\frac{N_1Q_1}{3\eta}-(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}-(\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}$$

$$\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_1}-\gamma_{12}\left[\frac{N_2Q_2}{3\eta}+\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta}-(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\right]=\frac{N_1Q_1}{3\eta}-(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}-(\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}$$

$$\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_1}=\frac{N_1Q_1}{3\eta}-(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}-(\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}+\gamma_{12}\frac{N_2Q_2}{3\eta}+\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta}$$
$$-\gamma_{12}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}$$

$$\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_1}=\frac{N_1Q_1}{3\eta}+\gamma_{12}\frac{N_2Q_2}{3\eta}-(\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+\left\{(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}-(\beta_{1\overline{2}})^{\frac{3}{2}}\right\}\frac{\overline{N_2}Q_1}{3\eta}-\gamma_{12}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}$$
$$+\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta} \tag{36}$$

5. Given that

$$\int dx\frac{(f(x))^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(f(x))^{\frac{5}{2}}-\frac{5}{7Q_2}f^{'}[xf^{'}-f]$$

$$\int dx\frac{(f(x))^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(f(x))^{\frac{5}{2}}-\frac{5}{7Q_2}\frac{df(x)}{dx}\left[x\frac{df(x)}{dx}-f\right] \tag{37}$$

6.

$$\int dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}-\frac{5}{7\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\left[x\frac{d\overline{f_2(x)}}{dx}-\overline{f_2(x)}\right]$$

$$\int_0^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{5}{7\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\left[x\frac{d\overline{f_2(x)}}{dx}-\overline{f_2(x)}\right]_{\overline{x_2}}$$

From eq^{n} (34),

$$\int_0^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{5}{7\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\frac{\overline{N_2Q_2}}{3\eta}$$

$$\int_0^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{5\overline{N_2}}{21\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \tag{38}$$

7.

$$\int_{\overline{x_2}}^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{\overline{x_2}}^{x_2}-\frac{5}{7Q_2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}^{x_2}\left[x\frac{df_2(x)}{dx}-f_2(x)\right]_{\overline{x_2}}^{x_2}$$

From eq^{n}s (34) & (35),

$$\int_{\overline{x_2}}^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{4}{7}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{5}{7Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}\left[\frac{N_2Q_2}{3\eta}+\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta}-(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\right]$$
$$+\frac{5}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\frac{\overline{N_2Q_2}}{3\eta} \tag{39}$$

8.

$$\int_{x_2}^{x_1}dx\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}=\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_2}^{x_1}-\frac{5}{7Q_1}\frac{df_1(x)}{dx}\Big|_{x_2}^{x_1}\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_2}^{x_1}$$

$$=\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}-\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5}{7Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_1}+\frac{5}{7Q_1}\frac{df_1(x)}{dx}\Big|_{x_2}\left[x\frac{df_1(x)}{dx}-f_1(x)\right]_{x_2}$$

$$\int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} = \frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} - \frac{4}{7}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5}{7Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}\left[\frac{N_1Q_1}{3\eta} + \gamma_{12}\frac{N_2Q_2}{3\eta} - (\gamma_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}\right.$$
$$\left. + \left\{(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - (\beta_{1\overline{2}})^{\frac{3}{2}}\right\}\frac{\overline{N_2}Q_1}{3\eta} - \gamma_{12}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta} + \gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta}\right] + \frac{5}{7Q_1}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$\left[\frac{N_2Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_2}{3\eta}\right] \tag{40}$$

Calculation of Potential Energy

$$U = -C\left[\sum_I \frac{(g_I N_I - 1)}{N_I} g_I K_{II} + \sum_{I\neq J} g_I g_J K_{IJ} + \tfrac{4}{3}\sum_{I,J} \overline{g_I} g_J K_{\overline{I}J}\right]$$

$$-\frac{U}{C} = \sum_I \frac{(g_I N_I - 1)}{N_I} g_I K_{II} + \sum_{I\neq J} g_I g_J K_{IJ} + \tfrac{4}{3}\sum_{I,J} \overline{g_I} g_J K_{\overline{I}J}$$

$$-\frac{U}{C} = \frac{(g_1N_1-1)}{N_1}g_1K_{11}+\frac{(g_2N_2-1)}{N_2}g_2K_{22}+g_1g_2K_{12}+g_2g_1K_{21}+\tfrac{4}{3}\overline{g_1}g_1K_{\overline{1}1}+\tfrac{4}{3}\overline{g_1}g_2K_{\overline{1}2}+\tfrac{4}{3}\overline{g_2}g_1K_{\overline{2}1}+\tfrac{4}{3}\overline{g_2}g_2K_{\overline{2}2}$$

Since, there are no light antiquarks, $\overline{f_1(x)} = 0$. It follows,

$$K_{\overline{1}1} = K_{\overline{1}2} = 0$$

$$\boxed{-\frac{U}{C} = \frac{(g_1N_1-1)}{N_1}g_1K_{11} + \frac{(g_2N_2-1)}{N_2}g_2K_{22} + \frac{(\overline{g_2}\overline{N_2}-1)}{\overline{N_2}}\overline{g_2}K_{\overline{2}\overline{2}} + g_1g_2K_{12} + g_2g_1K_{21} + \tfrac{4}{3}\overline{g_2}g_1K_{\overline{2}1} + \tfrac{4}{3}\overline{g_2}g_2K_{\overline{2}2}} \tag{41}$$

Our job is now to find the values of K_{11}, K_{21}, K_{12}, $K_{\overline{2}1}$, K_{22} & $K_{\overline{2}2}$ so that we can put these values in above equation and get the final expression of potential energy.

Calculation of K_{11}, K_{21}, K_{12}, $K_{\overline{2}1}$, K_{22}, $K_{\overline{2}2}$

Calculation of $K_{\overline{2}\overline{2}}$

$$K_{\overline{2}\overline{2}} = \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}dx \int_0^x \sqrt{x'}(\overline{f_2(x')})^{\frac{3}{2}}dx' + \int_0^{\overline{x_2}} \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}dx \int_x^{\overline{x_2}} \frac{(\overline{f_2(x')})^{\frac{3}{2}}}{\sqrt{x'}}dx'$$

$$= \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}dx\left[\frac{1}{\overline{\overline{Q_2}}}\left(x'\frac{d\overline{f_2(x')}}{dx'} - \overline{f_2(x')}\right)\Big|_x\right] + \int_0^{\overline{x_2}} \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}dx\frac{1}{\overline{\overline{Q_2}}}\int_x^{\overline{x_2}} \frac{d^2\overline{f_2(x')}}{dx'^2}dx'$$

$$= \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}dx \cdot \frac{1}{\overline{\overline{Q_2}}}\left(x'\frac{d\overline{f_2(x')}}{dx'} - \overline{f_2(x')}\right)\Big|_x + \frac{1}{\overline{\overline{Q_2}}}\int_0^{\overline{x_2}} \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}dx\left(\frac{d\overline{f_2(x')}}{dx'}\Big|_{\overline{x_2}} - \frac{d\overline{f_2(x')}}{dx'}\Big|_x\right)$$

$$= \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}dx \cdot \frac{1}{\overline{\overline{Q_2}}}\left(x'\frac{d\overline{f_2(x')}}{dx'} - \overline{f_2(x')}\right)\Big|_x + \frac{1}{\overline{\overline{Q_2}}}\frac{d\overline{f_2(x')}}{dx'}\Big|_{\overline{x_2}} \cdot \frac{\overline{N_2}}{3\eta} - \frac{1}{\overline{\overline{Q_2}}}\int_0^{\overline{x_2}} \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x')}}{dx'}\Big|_x dx$$

$$= \frac{1}{\overline{\overline{Q_2}}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx} - \frac{1}{\overline{\overline{Q_2}}}\int_0^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$= -\frac{1}{\overline{\overline{Q_2}}}\int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}dx + \frac{\overline{N_2}}{3\eta}\frac{1}{\overline{\overline{Q_2}}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$= -\frac{1}{\overline{\overline{Q_2}}}\left\{\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right\} + \frac{1}{\overline{\overline{Q_2}}}\frac{\overline{N_2}}{3\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$\boxed{K_{\overline{2}\overline{2}} = -\frac{4}{7\overline{\overline{Q_2}}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4\overline{N_2}}{7\eta\overline{\overline{Q_2}}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}} \tag{42}$$

Calculation of $K_{\overline{2}2}$

$$K_{\overline{2}2} = \int_0^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^x dx'\sqrt{x'}(f_2(x'))^{\frac{3}{2}} + \int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\int_x^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{43}$$

Substituting values from appendix

$$= \int_0^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right] + \int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left[\left\{\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\right.$$

$$\left. + \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}\right]$$

$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[\int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} x \frac{d\overline{f_2(x)}}{dx} - \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \overline{f_2(x)}\right] + \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \left[\left\{\frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2}\right\} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}}\right.$$

$$\left. - \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\Bigg|_{x_2}\right]$$

$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx (\overline{f_2(x)})^{\frac{3}{2}} \sqrt{x} \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \left\{\frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2}\right\} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}}$$

$$- \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2} \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \frac{df_2(x)}{dx}\Bigg|_{x_2}$$

$$= \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \left\{\frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2}\right\} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{1}{Q_2} \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2}\right\} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{1}{Q_2} \frac{\overline{N_2}}{3\eta} \frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Bigg|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}}\right]$$

$$= \frac{\overline{N_2}(\beta_{2\bar{2}})^{\frac{3}{2}}}{3\eta\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} - \frac{\overline{N_2}(\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right))}{3\eta Q_2} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2} \frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{4(\beta_{2\bar{2}})^{\frac{3}{2}}}{7\overline{Q_2}} \sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Bigg|_{\overline{x_2}} + \frac{5\overline{N_2}(\beta_{2\bar{2}})^{\frac{3}{2}}}{21\eta\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}}$$

$$= \frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2} + \frac{5(\beta_{2\bar{2}})^{\frac{3}{2}}}{7\overline{Q_2}}\right\} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} - \frac{4(\beta_{2\bar{2}})^{\frac{3}{2}}}{7\overline{Q_2}} \sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2} \frac{df_2(x)}{dx}\Bigg|_{x_2}$$

$$\boxed{K_{\bar{2}2} = -\frac{4}{7\overline{Q_2}}(\beta_{2\bar{2}})^{\frac{3}{2}} \sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2} + \frac{5(\beta_{2\bar{2}})^{\frac{3}{2}}}{7\overline{Q_2}}\right\} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2} \frac{df_2(x)}{dx}\Bigg|_{x_2}} \quad (44)$$

Appendix for $K_{\bar{2}2}$

1.

$$\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{Q_2} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

$$\frac{1}{\overline{Q_2}} \frac{d^2\overline{f_2(x)}}{dx^2} = \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

$$\int_0^{\overline{x_2}} \frac{1}{\overline{Q_2}} \frac{d^2\overline{f_2(x)}}{dx^2} = \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

$$\int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} = \frac{1}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} \quad (45)$$

2. The normalization condition for antiparticle is given as

$$\int_0^{x_{max}} dx \sqrt{x} (\overline{f^I(x)})^{\frac{3}{2}} = \frac{\overline{N^I}}{3\eta}$$

$$\int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta} \quad (46)$$

3.

$$\int_0^x dx \sqrt{x} (f_2(x))^{\frac{3}{2}}$$

In region $0 < x \leq \overline{x_2}$,

$$= (\beta_{2\bar{2}})^{\frac{3}{2}} \int_0^x dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}}$$

$$= (\beta_{2\bar{2}})^{\frac{3}{2}} \int_0^x dx \frac{x}{\overline{Q_2}} \frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^x dx x \frac{d^2\overline{f_2(x)}}{dx^2}$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_0^x$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right] \tag{47}$$

4.

$$\int_x^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$= \int_x^{\overline{x_2}} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$= (\beta_{2\bar{2}})^{\frac{3}{2}} \int_x^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_x^{\overline{x_2}} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{\overline{x_2}}^{x_2}$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_x + \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{\overline{x_2}}$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$= \left\{ \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2} \right\} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\Big|_{x_2} \tag{48}$$

Calculation of K_{22}

$$K_{22} = \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{49}$$

$$K_{22} = K_{22} \cdot 1 + K_{22} \cdot 2 \tag{50}$$

At first,

$$K_{22} \cdot 1 = \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} \tag{51}$$

$$= \int_0^{\overline{x_2}} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}$$

$$= \int_0^{\overline{x_2}} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left\{ x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right\} \right] + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\left\{ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \right\} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}} \right.$$
$$\left. + \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right] \right]$$

$$= \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx (\overline{f_2(x)})^{\frac{3}{2}} \sqrt{x} \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \left\{ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \right\} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$+ \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx} - \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

The two terms get cancelled with $K_{22} \cdot 2$

$$= \left\{ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \right\} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} - \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \frac{1}{Q_2} \left\{ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \right\} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}} \int_{\overline{x_2}}^{x_2} dx \frac{d^2 f_2(x)}{dx^2} - \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}} \left\{ \frac{4}{7} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right\}$$
$$- \frac{1}{Q_2} \left[\frac{4}{7} \sqrt{x} (f_2(x))^{\frac{5}{2}} \Big|_{x_2} - \frac{4}{7} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{5}{7Q_2} \frac{df_2(x)}{dx} \Big|_{x_2} \left\{ \frac{N_2 Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) \frac{\overline{N_2 Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta} \right\} \right.$$
$$\left. + \frac{5}{7Q_2} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2 \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \frac{\overline{N_2 Q_2}}{3\eta} \right]$$

$$= \frac{\overline{N_2 Q_2}}{3\eta Q_2} \left\{ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \right\} \frac{df_2(x)}{dx} \Big|_{\overline{x_2}}^{x_2} - \frac{4}{7} \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta} \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{4}{7Q_2} \sqrt{x} (f_2(x))^{\frac{5}{2}} \Big|_{x_2}$$
$$+ \frac{4}{7Q_2} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \frac{1}{Q_2} \frac{5}{7Q_2} \frac{N_2 Q_2}{3\eta} \frac{df_2(x)}{dx} \Big|_{x_2} + \frac{1}{Q_2} \frac{5}{7Q_2} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) \frac{\overline{N_2 Q_2}}{3\eta} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{1}{Q_2} \frac{5}{7Q_2} (\beta_{2\overline{2}})^{\frac{3}{2}}$$
$$\frac{\overline{N_2} Q_2}{3\eta} \frac{df_2(x)}{dx}_{x_2} - \frac{1}{Q_2} \frac{5}{7Q_2} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2 \frac{\overline{N_2 Q_2}}{3\eta} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}}$$

$$= \frac{\overline{N_2 Q_2}}{3\eta Q_2} \left\{ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \right\} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{\overline{N_2 Q_2}}{3\eta Q_2} \left\{ \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) (\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2}{Q_2} \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{4}{7} \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}}$$
$$+ \frac{5\overline{N_2}}{21\eta} \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{4}{7Q_2} \sqrt{x} (f_2(x))^{\frac{5}{2}} \Big|_{x_2} + \frac{4}{7Q_2} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \frac{5N_2}{21\eta Q_2} \frac{df_2(x)}{dx} \Big|_{x_2} + \frac{5\overline{N_2 Q_2}}{21\eta {Q_2}^2}$$
$$\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) \frac{df_2(x)}{dx}_{x_2} - \frac{5\overline{N_2}}{21\eta Q_2} (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{5\overline{N_2 Q_2}}{21\eta {Q_2}^2} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2 \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}}$$

$$= \frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{2\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{\overline{N_2Q_2}}{3\eta Q_2}\left\{\frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2}{Q_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\right\}$$

$$\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta}\frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2Q_2}}{3\eta Q_2}\left\{\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\right\}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{5N_2}{21\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \frac{4}{7}\left\{\frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}}{Q_2} - \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}}\right\}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}\left\{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{\overline{Q_2}}{Q_2^2} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} + \frac{5}{7\overline{Q_2}}(\beta_{2\overline{2}})^3\right.$$

$$\left. - \frac{5\overline{Q_2}}{7Q_2^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \left[\frac{5N_2}{21\eta Q_2} + \beta_{2\overline{2}}\frac{\overline{N_2}}{3\eta Q_2}\left\{(\beta_{2\overline{2}})^{\frac{1}{2}} - \frac{5}{7}(\beta_{2\overline{2}})^{\frac{1}{2}}\right\} - \frac{\overline{N_2Q_2}}{3\eta Q_2^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) + \frac{5\overline{N_2Q_2}}{21\eta Q_2^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right]\frac{df_2(x)}{dx}\Big|_{x_2} \tag{52}$$

At second,

$$K_{22}\cdot 2 = \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\int_x^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{53}$$

$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\int_x^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\int_x^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= (\beta_{2\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left\{\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right\}$$

$$+ \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\left\{\frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2}\frac{df_2(x)}{dx}\right\}$$

$$= \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx} + \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_2}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_2}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_2}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}$$

The two terms get cancelled with $K_{22}\cdot 1$

$$= \frac{(\beta_{2\overline{2}})^3}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_2}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{Q_2}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_2}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$= (\beta_{2\overline{2}})^3\frac{\overline{N_2}}{3\eta\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + (\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}}{3\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}}{3\eta Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_2}\frac{N_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- (\beta_{2\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}}{3\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

Therefore,

$$K_{22} = K_{22}\cdot 1 + K_{22}\cdot 2$$

$$K_{22} = \frac{4}{7}\Bigg\{ \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}}{Q_2} - \frac{(\beta_{2\bar{2}})^3}{\overline{Q_2}} \Bigg\}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}\Bigg\{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)^2\frac{\overline{Q_2}}{Q_2^2} - \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2} + \frac{5}{7\overline{Q_2}}(\beta_{2\bar{2}})^3$$
$$- \frac{5\overline{Q_2}}{7Q_2^2}\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)^2\Bigg\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \Bigg[\frac{5N_2}{21\eta Q_2} + \beta_{2\bar{2}}\frac{\overline{N_2}}{3\eta Q_2}\Bigg\{(\beta_{2\bar{2}})^{\frac{1}{2}} - \frac{5}{7}(\beta_{2\bar{2}})^{\frac{1}{2}}\Bigg\} - \frac{\overline{N_2Q_2}}{3\eta Q_2^2}\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right) + \frac{5\overline{N_2Q_2}}{21\eta Q_2^2}$$
$$\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)\Bigg]\frac{df_2(x)}{dx}\Big|_{x_2} + (\beta_{2\bar{2}})^3\frac{\overline{N_2}}{3\eta\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)(\beta_{2\bar{2}})^{\frac{3}{2}}\frac{\overline{N_2}}{3\eta Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_2}\frac{N_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2} \tag{54}$$

Appendix for K_{22}

1.

$$\int_0^x dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$

In region $0 < x \leq \overline{x_2}$,

$$= (\beta_{2\bar{2}})^{\frac{3}{2}}\int_0^x dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$
$$= (\beta_{2\bar{2}})^{\frac{3}{2}}\int_0^x dx\frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2}$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_0^x dxx\frac{d^2\overline{f_2(x)}}{dx^2}$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right] \tag{55}$$

2.

$$\int_0^x dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$

In region $\overline{x_2} < x \leq x_2$,

$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} + \int_{\overline{x_2}}^x dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$
$$= (\beta_{2\bar{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} + \int_{\overline{x_2}}^x dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$
$$= (\beta_{2\bar{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2} + \int_{\overline{x_2}}^x dx\frac{x}{Q_2}\frac{d^2f_2(x)}{dx^2}$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_0^{\overline{x_2}} dxx\frac{d^2\overline{f_2(x)}}{dx^2} + \frac{1}{Q_2}\int_{\overline{x_2}}^x dxx\frac{d^2f_2(x)}{dx^2}$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}} + \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}^x$$
$$= \frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}} + \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right] - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}$$
$$= \left\{\frac{(\beta_{2\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2}\right\}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}} + \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right] \tag{56}$$

3.

$$\int_x^{x_2} dx\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$

In region $0 < x \leq \overline{x_2}$,

$$= \int_x^{\overline{x_2}} dx\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \int_{\overline{x_2}}^{x_2} dx\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$= (\beta_{2\bar{2}})^{\frac{3}{2}}\int_x^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + \int_{\overline{x_2}}^{x_2} dx\frac{1}{Q_2}\frac{d^2f_2(x)}{dx^2}$$

$$= (\beta_{2\overline{2}})^{\frac{3}{2}} \int_x^{\overline{x_2}} dx \frac{1}{\overline{Q_2}} \frac{d^2\overline{f_2(x)}}{dx^2} + \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_x^{\overline{x_2}} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\bigg|_{\overline{x_2}}^{x_2}$$

$$= \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \tag{57}$$

4.

$$\int_x^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$

In region $\overline{x_2} < x \leq x_2$,

$$= \int_x^{x_2} dx \frac{1}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \frac{df_2(x)}{dx}\bigg|_x^{x_2}$$

$$= \frac{1}{Q_2} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{1}{Q_2} \frac{df_2(x)}{dx} \tag{58}$$

5.

$$\int_{\overline{x_2}}^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}}$$

$$= \int_{\overline{x_2}}^{x_2} dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx x \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}}^{x_2}$$

$$= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}}$$

$$= \frac{1}{Q_2} \left[\frac{N_2 Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) \frac{\overline{N_2 Q_2}}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2} Q_2}{3\eta} \right] - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \frac{\overline{N_2 Q_2}}{3\eta}$$

$$= \frac{N_2}{3\eta} + \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{3\eta} \frac{\overline{N_2 Q_2}}{Q_2} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \frac{\overline{N_2 Q_2}}{3\eta}$$

$$= \frac{N_2}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta} \tag{59}$$

Calculation of $K_{\overline{2}1}$

$$K_{\overline{2}1} = \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{60}$$

$$K_{\overline{2}1} = K_{\overline{2}1} \cdot 1 + K_{\overline{2}1} \cdot 2 \tag{61}$$

At first,

$$K_{\overline{2}1} \cdot 1 = \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \int_0^x dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} \tag{62}$$

$$= \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \cdot \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left\{ x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right\}$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \cdot x \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx (\overline{f_2(x)})^{\frac{3}{2}} \sqrt{x} \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

The first term get cancelled with $K_{\overline{2}1} \cdot 2$

$$= - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[\frac{4}{7} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right]$$

$$= - \frac{4}{7\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \tag{63}$$

At second,

$$K_{\overline{2}1} \cdot 2 = \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{64}$$

$$= \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \left[\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{(\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)}{Q_2} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{1}{Q_1} \frac{df_1(x)}{dx} \Big|_{x_1} \right.$$

$$\left. - \frac{\gamma_{12}}{Q_1} \frac{df_2(x)}{dx} \Big|_{x_2} \right]$$

The second term in the bracket get cancelled with $K_{\overline{2}1} \cdot 1$

$$= \frac{\overline{N_2}}{3\eta\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{\overline{N_2}(\gamma_{12})^{\frac{3}{2}}}{3\eta Q_2} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_1} \frac{df_1(x)}{dx} \Big|_{x_1} - \frac{\overline{N_2}}{3\eta Q_1} \gamma_{12} \frac{df_2(x)}{dx} \Big|_{x_2} \tag{65}$$

Therefore,

$$K_{\overline{2}1} = K_{\overline{2}1} \cdot 1 + K_{\overline{2}1} \cdot 2$$

$$= - \frac{4}{7\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{\overline{N_2}(\gamma_{12})^{\frac{3}{2}}}{3\eta Q_2} \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$- \frac{\overline{N_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_1} \frac{df_1(x)}{dx} \Big|_{x_1} - \frac{\overline{N_2}}{3\eta Q_1} \gamma_{12} \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$= - \frac{4}{7\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \left\{ \frac{5\overline{N_2}}{21\eta\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{\overline{N_2}}{3\eta\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \left\{ \frac{\overline{N_2}(\gamma_{12})^{\frac{3}{2}}}{3\eta Q_2} \right.$$

$$\left. - \frac{\overline{N_2}\gamma_{12}}{3\eta Q_1} \right\} \frac{df_2(x)}{dx} \Big|_{x_2} + \frac{\overline{N_2}}{3\eta Q_1} \frac{df_1(x)}{dx} \Big|_{x_1}$$

$$\boxed{K_{\overline{2}1} = - \frac{4}{7\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta} \left\{ \frac{5}{7\overline{Q_2}} (\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{(\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)}{Q_2} \right\} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta} \left\{ \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \right.$$

$$\left. - \frac{\gamma_{12}}{Q_1} \right\} \frac{df_2(x)}{dx} \Big|_{x_2} + \frac{\overline{N_2}}{3\eta Q_1} \frac{df_1(x)}{dx} \Big|_{x_1}} \tag{66}$$

Appendix for $K_{\overline{2}1}$

1.

$$\int_0^x dx\sqrt{x}(f_1(x))^{\frac{3}{2}}$$

In region $0 < x \leq \overline{x_2}$,

$$= (\beta_{1\overline{2}})^{\frac{3}{2}} \int_0^x dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}} \int_0^x dx \frac{x}{\overline{Q_2}} \frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^x dxx \frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right] \tag{67}$$

2.

$$\int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= \int_x^{\overline{x_2}} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}} \int_x^{\overline{x_2}} dx' \frac{(\overline{f_2(x')})^{\frac{3}{2}}}{\sqrt{x'}} + (\gamma_{12})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_x^{\overline{x_2}} dx \frac{d^2\overline{f_2(x)}}{dx^2} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{d^2 f_2(x)}{dx^2} + \frac{1}{Q_1} \int_{x_2}^{x_1} dx \frac{d^2 f_1(x)}{dx^2}$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{1}{Q_2}(\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{1}{Q_1}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{1}{Q_1}\gamma_{12}\frac{df_2(x)}{dx}\bigg|_{x_2} \tag{68}$$

Calculation of K_{12}

$$K_{12} = \int_0^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_0^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{69}$$

$$K_{12} = K_{12} \cdot 1 + K_{12} \cdot 2 \tag{70}$$

At first,

$$K_{12} \cdot 1 = \int_0^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} \tag{71}$$

$$K_{12}\cdot 1 = \int_0^{\overline{x_2}} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}$$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \left\{ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right] \right\} + (\gamma_{12})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\int_0^{\overline{x_2}} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} \right.$$
$$\left. + \int_{\overline{x_2}}^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} \right] + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\int_0^{\overline{x_2}} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_{x_2}^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} \right]$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[\int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \cdot x \frac{d\overline{f_2(x)}}{dx} - \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \cdot \overline{f_2(x)} \right] + (\gamma_{12})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{\overline{N_2}}{3\eta} (\beta_{2\overline{2}})^{\frac{3}{2}} \right.$$
$$\left. + \frac{1}{Q_2} \left(x \frac{df_2(x)}{dx} - f_2(x) \right) - \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{\overline{N_2 Q_2}}{3\eta Q_2} \right] + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{\overline{N_2}}{3\eta} (\beta_{2\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta} - (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{\overline{N_2}}{3\eta} \right]$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \cdot x \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \frac{\overline{N_2}}{3\eta} (\gamma_{12})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$+ \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \cdot x \frac{df_2(x)}{dx} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$+ \frac{N_2}{3\eta} \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

The first and fourth term gets cancelled with $K_{12} \cdot 2$

$$= \frac{\overline{N_2}}{3\eta} (\gamma_{12})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$
$$- \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta} \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$= \frac{\overline{N_2}}{3\eta} (\gamma_{12})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}} \left[\frac{1}{Q_2} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)}{Q_2} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right] - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[\frac{4}{7} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right]$$
$$- \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left[\frac{4}{7} \sqrt{x} (f_2(x))^{\frac{5}{2}} \Big|_{x_2} - \frac{4}{7} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)^{\frac{5}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{5}{7Q_2} \frac{df_2(x)}{dx} \Big|_{x_2} \left\{ \frac{N_2 Q_2}{3\eta} + \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{\overline{N_2 Q_2}}{3\eta} - \frac{\overline{N_2} Q_2}{3\eta} (\beta_{2\overline{2}})^{\frac{3}{2}} \right\} \right.$$
$$\left. + \frac{5}{7Q_2} \frac{\overline{N_2 Q_2}}{3\eta} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)^2 \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right] - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \left[\frac{1}{Q_2} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)}{Q_2} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right] + \frac{N_2}{3\eta Q_1} \frac{df_1(x)}{dx} \Big|_{x_1}$$
$$- \frac{N_2}{3\eta Q_1} \gamma_{12} \frac{df_2(x)}{dx} \Big|_{x_2} \tag{72}$$

At second,

$$K_{12} \cdot 2 = \int_0^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{73}$$

$$= \int_0^{\overline{x_2}} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\left[\int_x^{\overline{x_2}} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}\right] + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\left[\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_x^{\overline{x_2}} dx'\frac{d^2\overline{f_2(x')}}{dx'^2} + \frac{1}{Q_2}\int_{\overline{x_2}}^{x_2} dx'\frac{d^2f_2(x')}{dx'^2}\right] + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\left[\frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2}\frac{df_2(x)}{dx}\right]$$

$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\left[\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}\right] + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}$$
$$\left[\frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2}\frac{df_2(x)}{dx}\right]$$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left[\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}\right] - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}$$
$$+ \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}$$

The second term and fourth term gets cancelled with $K_{12}\cdot 1$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left[\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{\overline{x_2}}\right] + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}}{3\eta}\left[\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[\frac{1}{Q_2}\left(x\frac{df_2(x)}{dx} - f_2(x)\right)_{x_2}\right.$$
$$\left. - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\left(x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right)_{\overline{x_2}}\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \cdots + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[\frac{1}{Q_2}\left\{\frac{N_2Q_2}{3\eta} + \frac{\overline{N_2Q_2}}{3\eta}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}Q_2}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\right\} - \frac{\overline{N_2Q_2}}{3\eta Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \cdots + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[\frac{N_2}{3\eta} + \frac{\overline{N_2Q_2}}{3\eta Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right]\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \cdots + \frac{N_2}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}}{3\eta}\left[\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] + \frac{N_2}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} \tag{74}$$

Therefore, $K_{12} = K_{12}\cdot 1 + K_{12}\cdot 2$

$$= \frac{\overline{N_2}}{3\eta}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\left[\frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] - \frac{1}{\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\left[\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right]$$
$$- \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5}{7Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}\left\{\frac{N_2Q_2}{3\eta} + \frac{\overline{N_2Q_2}}{3\eta}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}Q_2}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\right\}\right.$$
$$\left. + \frac{5}{7Q_2}\frac{\overline{N_2Q_2}}{3\eta}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] - \frac{\overline{N_2Q_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\left[\frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$
$$- \frac{N_2}{3\eta Q_1}\gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}}\left[\frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] + \frac{N_2}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$- \frac{\overline{N_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$
\begin{aligned}
&= -\frac{\overline{N_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&- \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{5N_2}{21\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} \\
&+ \frac{5\overline{N_2Q_2}}{21\eta {Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2Q_2}}{21\eta {Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&- \frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2}{3\eta Q_1}\gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2} \\
&+ \frac{\overline{N_2}}{3\eta\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}
\end{aligned}
$$

$$
\boxed{
\begin{aligned}
K_{12} = &\frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} \\
&+ \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta {Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&+ \frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&+ \frac{5N_2}{21\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2Q_2}}{21\eta {Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} \\
&- \frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2}{3\eta Q_1}\gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} \\
&+ \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
} \tag{75}
$$

Appendix for K_{12}

1.

$$
\begin{aligned}
\int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_{x}^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} &= \int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\left(-\int_{x_2}^{x} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}\right) \\
&= 0
\end{aligned} \tag{76}
$$

This is because $f_2(x)$ doesnot exist in region $> x_2$.

2.

$$
\begin{aligned}
&\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \\
&= \frac{1}{Q_2}\int_{\overline{x_2}}^{x_2} dx x\frac{d^2f_2(x)}{dx^2} \\
&= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}^{x_2} \\
&= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}} \\
&= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}
\end{aligned} \tag{77}
$$

3.

$$
\int_0^x dx\sqrt{x}(f_2(x))^{\frac{3}{2}}
$$

In region $0 < x \leq \overline{x_2}$,

$$
= \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right] \tag{78}
$$

4. The normalization condition gives

$$\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta}$$
$$\frac{1}{(\beta_{2\overline{2}})^{\frac{3}{2}}}\int_0^{\overline{x_2}} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta}$$
$$\int_0^{\overline{x_2}} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}} \tag{79}$$

5.

$$\int_{\overline{x_2}}^{x} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$
$$= \int_{\overline{x_2}}^{x} dx\frac{x}{Q_2}\frac{d^2f_2(x)}{dx^2}$$
$$= \frac{1}{Q_2}\int_{\overline{x_2}}^{x} dxx\frac{d^2f_2(x)}{dx^2}$$
$$= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}^{x}$$
$$= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right] - \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}$$
$$= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right] - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}$$
$$= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right] - \frac{\overline{N_2Q_2}}{3\eta Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) \tag{80}$$

6.

$$\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$
$$= \int_{\overline{x_2}}^{x_2} dx\frac{x}{Q_2}\frac{d^2f_2(x)}{dx^2}$$
$$= \frac{1}{Q_2}\int_{\overline{x_2}}^{x_2} dxx\frac{d^2f_2(x)}{dx^2}$$
$$= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}^{x_2}$$
$$= \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \frac{1}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}$$
$$= \frac{1}{Q_2}\left[\frac{N_2Q_2}{3\eta} + \frac{\overline{N_2Q_2}}{3\eta}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}Q_2}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\right] - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}$$
$$= \frac{N_2}{3\eta} + \frac{\overline{N_2Q_2}}{3\eta Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)$$
$$= \frac{N_2}{3\eta} - \frac{\overline{N_2}}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}} \tag{81}$$

Calculation of K_{21}

$$K_{21} = \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{82}$$

$$K_{21} = K_{21} \cdot 1 + K_{21} \cdot 2 \tag{83}$$

At first,

$$K_{21} \cdot 1 = \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} \tag{84}$$

$$= \int_0^{\overline{x_2}} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}}$$

$$= (\beta_{2\overline{2}})^{\frac{3}{2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \left\{ \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left(x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right) \right\} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\int_0^{\overline{x_2}} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} \right]$$

$$= \frac{(\beta_{2\overline{2}})^{\frac{3}{2}} (\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[\int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \cdot x \frac{d\overline{f_2(x)}}{dx} - \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} \right] + \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left(x \frac{df_2(x)}{dx} - f_2(x) \right) \right.$$

$$\left. - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \right]$$

$$= \frac{(\beta_{2\overline{2}})^{\frac{3}{2}} (\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx (\overline{f_2(x)})^{\frac{3}{2}} \sqrt{x} \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}} (\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$+ \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \int_{\overline{x_2}}^{x_2} dx (f_2(x))^{\frac{3}{2}} \sqrt{x} \frac{df_2(x)}{dx} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$

The first term and fourth term gets cancelled with $K_{21} \cdot 2$

$$= \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}} (\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)$$

$$\int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$= \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{1}{Q_2} \frac{d^2 f_2(x)}{dx^2} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}} (\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[\frac{4}{7} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right] - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left[\frac{4}{7} \sqrt{x} (f_2(x))^{\frac{5}{2}} \Big|_{x_2} \right.$$

$$\left. - \frac{4}{7} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)^{\frac{5}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{5}{7Q_2} \frac{df_2(x)}{dx} \Big|_{x_2} \left\{ \frac{N_2 Q_2}{3\eta} + \frac{\overline{N_2 Q_2}}{3\eta} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) - \frac{\overline{N_2} Q_2}{3\eta} (\beta_{2\overline{2}})^{\frac{3}{2}} \right\} + \frac{5}{7Q_2} \frac{\overline{N_2 Q_2}}{3\eta} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)^2 \right.$$

$$\left. \frac{d\overline{f_2(x)}}{dx}_{\overline{x_2}} \right] - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \int_{\overline{x_2}}^{x_2} dx \frac{1}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{\overline{N_2}}{3\eta Q_2} (\beta_{1\overline{2}})^{\frac{3}{2}} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2} (\beta_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}} (\beta_{2\overline{2}})^{\frac{3}{2}} (\beta_{1\overline{2}})^{\frac{3}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta \overline{Q_2}} (\beta_{2\overline{2}})^{\frac{3}{2}}$$

$$(\beta_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}_{\overline{x_2}} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left[\frac{4}{7} \sqrt{x} (f_2(x))^{\frac{5}{2}} \Big|_{x_2} - \frac{4}{7} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)^{\frac{5}{2}} \sqrt{x} (\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{5N_2}{21\eta} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{5\overline{N_2 Q_2}}{21\eta Q_2} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{df_2(x)}{dx} \Big|_{x_2} \right.$$

$$\left. + \frac{5\overline{N_2}}{21\eta} (\beta_{2\overline{2}})^{\frac{3}{2}} \frac{df_2(x)}{dx} \Big|_{x_2} + \frac{5\overline{N_2 Q_2}}{21\eta Q_2} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)^2 \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right] - \frac{\overline{N_2 Q_2}}{3\eta {Q_2}^2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$+ \frac{\overline{N_2 Q_2}}{3\eta {Q_2}^2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right)^2 \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}}$$

$$= \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{3}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{3}{2}}$$
$$(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}_{\overline{x_2}} - \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{5N_2}{21\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$+ \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$- \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \tag{85}$$

$$= \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{3}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}$$
$$+ \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{3}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$+ \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5N_2}{21\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$- \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} \tag{86}$$

At second,

$$K_{21}\cdot 2 = \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{87}$$
$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$
$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\left\{\int_x^{\overline{x_2}} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}\right\} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$
$$\left\{\int_x^{x_2} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}\right\}$$
$$= (\beta_{2\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left\{(\beta_{1\overline{2}})^{\frac{3}{2}}\int_x^{\overline{x_2}} dx'\frac{(\overline{f_2(x')})^{\frac{3}{2}}}{\sqrt{x'}} + (\gamma_{12})^{\frac{3}{2}}\int_{\overline{x_2}}^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}\right\}$$
$$+ \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\left\{(\gamma_{12})^{\frac{3}{2}}\int_x^{x_2} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}\right\}$$
$$= (\beta_{2\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left[\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_x^{\overline{x_2}} dx'\frac{d^2\overline{f_2(x')}}{dx'^2} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^{x_2} dx'\frac{d^2f_2(x')}{dx'^2} + \frac{1}{Q_1}\int_{x_2}^{x_1} dx'\frac{d^2f_1(x')}{dx'^2}\right]$$
$$+ \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x'))^{\frac{3}{2}}\left[\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_x^{x_2} dx'\frac{d^2f_2(x')}{dx'^2} + \frac{1}{Q_1}\int_{x_2}^{x_1} dx'\frac{d^2f_1(x')}{dx'^2}\right]$$

$$= (\beta_{2\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left[\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right.$$
$$\left. + \frac{1}{Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{\gamma_{12}}{Q_1}\frac{df_2(x)}{dx}\Big|_{x_2}\right] + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\left[\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\frac{df_2(x)}{dx} + \frac{1}{Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{\gamma_{12}}{Q_1}\frac{df_2(x)}{dx}\Big|_{x_2}\right.$$

$$= \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}$$
$$+ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_2}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$+ \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_1}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}}{Q_1}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$- \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx} + \frac{1}{Q_1}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{\gamma_{12}}{Q_1}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

The second term and eighth term gets cancelled with $K_{21}\cdot 1$

$$= \frac{\overline{N_2}}{3\eta\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{3}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_1}$$
$$(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}_{x_1} - \frac{\overline{N_2}}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$
$$- \frac{\overline{N_2}}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2}{3\eta Q_1}\gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\left\{\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2}{3\eta}\gamma_{12}\left\{\frac{(\gamma_{12})^{\frac{1}{2}}}{Q_2} - \frac{1}{Q_1}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} \quad (88)$$

Therefore, $K_{21} = K_{21}\cdot 1 + K_{21}\cdot 2$

$$= \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{3}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}$$
$$+ \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{3}{2}}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$+ \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5N_2}{21\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$- \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\left\{\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$+ \frac{N_2}{3\eta}\gamma_{12}\left\{\frac{(\gamma_{12})^{\frac{1}{2}}}{Q_2} - \frac{1}{Q_1}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$

$$\begin{aligned}K_{12} = {} & \frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_2}(\gamma_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} \\ & + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\ & + \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta\overline{Q_2}}(\beta_{1\overline{2}})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\ & + \frac{5N_2}{21\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} \\ & - \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2}{3\eta Q_1}\gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} \\ & + \frac{N_2}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}\end{aligned} \quad (89)$$

Appendix for K_{21}

1.

$$\int_0^x dx\sqrt{x}(f_1(x))^{\frac{3}{2}}$$

In region $0 < x \leq \overline{x_2}$,

$$= (\beta_{1\bar{2}})^{\frac{3}{2}} \int_0^x dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

$$= (\beta_{1\bar{2}})^{\frac{3}{2}} \int_0^x dx \frac{x}{\overline{Q_2}} \frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{1\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^x dxx \frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{1\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_0^x$$

$$= \frac{(\beta_{1\bar{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right] \tag{90}$$

2. The normalization condition gives

$$\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta}$$

$$\frac{1}{(\beta_{1\bar{2}})^{\frac{3}{2}}} \int_0^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta}$$

$$\int_0^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta}(\beta_{1\bar{2}})^{\frac{3}{2}} \tag{91}$$

3.

$$\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$

$$= \int_{\overline{x_2}}^{x_2} dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dxx \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{1}{Q_2} \left[x\frac{df_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}}^{x_2}$$

$$= \frac{1}{Q_2} \left[x\frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \frac{\left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)}{Q_2} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}}$$

$$= \frac{1}{Q_2} \left[\frac{N_2 Q_2}{3\eta} + \frac{\overline{N_2 Q_2}}{3\eta} \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2} Q_2}{3\eta} (\beta_{2\bar{2}})^{\frac{3}{2}} \right] - \frac{\overline{N_2 Q_2}}{3\eta Q_2} \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)$$

$$= \frac{N_2}{3\eta} + \frac{\overline{N_2 Q_2}}{3\eta Q_2} \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}}{3\eta} (\beta_{2\bar{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_2} \left(\frac{\beta_{1\bar{2}}}{\gamma_{12}}\right)$$

$$= \frac{N_2}{3\eta} - \frac{\overline{N_2}}{3\eta} (\beta_{2\bar{2}})^{\frac{3}{2}} \tag{92}$$

Calculation of K_{11}

$$K_{11} = \int_0^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_0^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_1} dx \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{93}$$

$$K_{11} = K_{11} \cdot 1 + K_{11} \cdot 2 \tag{94}$$

At first,

$$K_{11} \cdot 1 = \int_0^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} \tag{95}$$

$$\begin{aligned}
&= \int_0^{\overline{x_2}} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} \\
&= (\beta_{1\overline{2}})^{\frac{3}{2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left\{ x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right\} \right] + (\gamma_{12})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^{\frac{3}{2}} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \right. \\
&\quad \left. + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2} \left\{ x \frac{df_2(x)}{dx} - f_2(x) \right\} \right] + \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta} (\gamma_{12})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta Q_1} \gamma_{12} - \frac{\overline{N_2}}{3\eta} (\gamma_{12})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}} \right. \\
&\quad \left. - \frac{\overline{N_2 Q_2}}{3\eta Q_1} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \gamma_{12} + \frac{\overline{N_2} Q_2}{3\eta Q_1} (\beta_{2\overline{2}})^{\frac{3}{2}} \gamma_{12} + \frac{1}{Q_1} \left\{ x \frac{df_1(x)}{dx} - f_1(x) \right\} \right] \\
&= \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} - \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} (\gamma_{12})^{\frac{3}{2}} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \\
&\quad - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^3 \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \frac{(\gamma_{12})^3}{Q_2} \int_{\overline{x_2}}^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx} - \frac{(\gamma_{12})^3}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} \\
&\quad + \left\{ \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta} (\gamma_{12})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta Q_1} \gamma_{12} - \frac{\overline{N_2}}{3\eta} (\gamma_{12})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_1} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \gamma_{12} + \frac{\overline{N_2} Q_2}{3\eta Q_1} \gamma_{12} (\beta_{2\overline{2}})^{\frac{3}{2}} \right\} \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \\
&\quad + \frac{1}{Q_1} \int_{x_2}^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \frac{df_1(x)}{dx} - \frac{1}{Q_1} \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} \\
&= \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} + \frac{(\gamma_{12})^3}{Q_2} \int_{\overline{x_2}}^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx} + \frac{1}{Q_1} \int_{x_2}^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \frac{df_1(x)}{dx} \\
&\quad - \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{12})^3}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{1}{Q_1} \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \left\{ \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} (\gamma_{12})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^3 \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \right\} \\
&\quad \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \left\{ \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta} (\gamma_{12})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta Q_1} \gamma_{12} - \frac{\overline{N_2}}{3\eta} (\gamma_{12})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_1} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \gamma_{12} + \frac{\overline{N_2} Q_2}{3\eta Q_1} (\beta_{2\overline{2}})^{\frac{3}{2}} \gamma_{12} \right\} \\
&\quad \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}
\end{aligned}$$

The first, second, and third terms get canceled with $K_{11} \cdot 2$,

$$\begin{aligned}
&= -\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{12})^3}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{1}{Q_1} \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \left\{ \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} (\gamma_{12})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_2} (\gamma_{12})^3 \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \right\} \\
&\quad \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \left\{ \frac{\overline{N_2}}{3\eta} (\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta} (\gamma_{12})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta Q_1} \gamma_{12} - \frac{\overline{N_2}}{3\eta} (\gamma_{12})^{\frac{3}{2}} (\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_1} \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}} \right) \gamma_{12} + \frac{\overline{N_2} Q_2}{3\eta Q_1} (\beta_{2\overline{2}})^{\frac{3}{2}} \gamma_{12} \right\} \\
&\quad \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}
\end{aligned} \tag{96}$$

For convenience, above equation is evaluated in three sections separately.

- Section 1

$$-\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{12})^3}{Q_2} \int_{\overline{x_2}}^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{1}{Q_1} \int_{x_2}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} \tag{97}$$

Substituting values of above integration from appendix

$$
\begin{aligned}
&= -\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\left[\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5\overline{N_2}}{21\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] - \frac{(\gamma_{12})^3}{Q_2}\left[\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5}{7Q_2}\left\{\frac{N_2Q_2}{3\eta}\right.\right.\\
&\quad\left.\left. + \frac{\overline{N_2Q_2}}{3\eta}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}Q_2}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5}{7Q_2}\frac{\overline{N_2Q_2}}{3\eta}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right] - \frac{1}{Q_1}\left[\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}\right.\\
&\quad - \frac{4}{7}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5}{7Q_1}\left[\frac{N_1Q_1}{3\eta} + \frac{N_2Q_2}{3\eta}\gamma_{12} - \frac{N_2Q_1}{3\eta}(\gamma_{12})^{\frac{3}{2}} + \frac{\overline{N_2}Q_1}{3\eta}\left\{(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - (\beta_{1\overline{2}})^{\frac{3}{2}}\right\}\right.\\
&\quad\left.\left. - \frac{\overline{N_2}Q_2}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12} + \frac{\overline{N_2Q_2}}{3\eta}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right]\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5}{7Q_1}(\gamma_{12})^2\left[\frac{N_2Q_2}{3\eta} + \frac{\overline{N_2Q_2}}{3\eta}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}Q_2}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\right]\frac{df_2(x)}{dx}\Big|_{x_2}\right]\\
&= -\frac{4}{7}\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta}\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{4}{7}\frac{(\gamma_{12})^3}{Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}\\
&\quad + \frac{5}{7Q_2}\frac{N_2Q_2}{3\eta}\frac{(\gamma_{12})^3}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5}{7Q_2}\frac{\overline{N_2Q_2}}{3\eta Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)(\gamma_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5}{7Q_2}\frac{\overline{N_2}Q_2}{3\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}\\
&\quad - \frac{5}{7Q_2}\frac{\overline{N_2Q_2}}{3\eta Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \frac{4}{7Q_1}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_1Q_1}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1}\\
&\quad + \frac{5}{7Q_1}\frac{N_2Q_2}{3\eta Q_1}\gamma_{12}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2Q_1}{3\eta}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{1}{Q_1}\frac{5}{7Q_1}\frac{\overline{N_2}Q_1}{3\eta}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}\\
&\quad - \frac{1}{Q_1}\frac{5}{7Q_1}\frac{\overline{N_2}Q_1}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{1}{Q_1}\frac{5}{7Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{\overline{N_2}Q_2}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{1}{Q_1}\frac{5}{7Q_1}\frac{\overline{N_2Q_2}}{3\eta}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\Big|_{x_1}\\
&\quad - \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2Q_2}{3\eta}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{1}{Q_1}\frac{5}{7Q_1}\frac{\overline{N_2Q_2}}{3\eta}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{Q_1}\frac{5}{7Q_1}\frac{\overline{N_2}Q_2}{3\eta}(\gamma_{12})^2(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}\\
&= \frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{1\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7Q_1}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}(\gamma_{12})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}\\
&\quad - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5N_2}{21\eta Q_2}(\gamma_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}\\
&\quad + \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5N_2Q_2}{21\eta{Q_1}^2}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}(\gamma_{12})^2\\
&\quad \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2}Q_2}{21\eta{Q_1}^2}(\gamma_{12})^2(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5N_2Q_2}{21\eta{Q_1}^2}\gamma_{12}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5N_2}{21\eta Q_1}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}\\
&\quad + \frac{5\overline{N_2}}{21\eta Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5\overline{N_2}}{21\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5\overline{N_2}Q_2}{21\eta{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
\tag{98}
$$

- Section 2

$$
\left\{\frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right\}\int_{\overline{x_2}}^{x_2} dx\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{99}
$$

$$
\begin{aligned}
&= \frac{1}{Q_2}\left\{\frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right\}\int_{\overline{x_2}}^{x_2} dx\frac{d^2f_2(x)}{dx^2}\\
&= \frac{1}{Q_2}\left\{\frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right\}\left[\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{df_2(x)}{dx}\Big|_{\overline{x_2}}\right]\\
&= \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\\
&\quad + \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}
\end{aligned}
\tag{100}
$$

- Section 3

$$\left\{\frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}}+\frac{N_2}{3\eta}(\gamma_{12})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta Q_1}\gamma_{12}-\frac{\overline{N_2}}{3\eta}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta Q_1}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}+\frac{\overline{N_2}Q_2}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\}\int_{x_2}^{x_1}dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{101}$$

$$=\frac{1}{Q_1}\left\{\frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}}+\frac{N_2}{3\eta}(\gamma_{12})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta Q_1}\gamma_{12}-\frac{\overline{N_2}}{3\eta}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta Q_1}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}+\frac{\overline{N_2}Q_2}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\}\int_{x_2}^{x_1}dx\frac{d^2f_1(x)}{dx^2}$$

$$=\frac{1}{Q_1}\left\{\frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}}+\frac{N_2}{3\eta}(\gamma_{12})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta Q_1}\gamma_{12}-\frac{\overline{N_2}}{3\eta}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta Q_1}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}+\frac{\overline{N_2}Q_2}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\}\left[\left.\frac{df_1(x)}{dx}\right|_{x_1}-\left.\frac{df_1(x)}{dx}\right|_{x_2}\right]$$

$$\begin{aligned}&=\left\{\frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}+\frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta {Q_1}^2}\gamma_{12}-\frac{\overline{N_2}}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}+\frac{\overline{N_2}Q_2}{3\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\}\left.\frac{df_1(x)}{dx}\right|_{x_1}\\&-\left\{\frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}+\frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}}-\frac{N_2Q_2}{3\eta {Q_1}^2}(\gamma_{12})^2-\frac{\overline{N_2}}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right.\\&\left.+\frac{\overline{N_2}Q_2}{3\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^2\right\}\left.\frac{df_2(x)}{dx}\right|_{x_2}\end{aligned} \tag{102}$$

Adding all three sections

$$\begin{aligned}&=\frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{4}{7\overline{Q_2}}(\beta_{1\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}+\frac{4}{7Q_1}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{4}{7Q_2}(\gamma_{12})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}\\&-\frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}+\frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{5\overline{N_2Q_2}}{21\eta {Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}+\frac{5N_2}{21\eta Q_2}(\gamma_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}\\&+\frac{5\overline{N_2Q_2}}{21\eta {Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5N_2Q_2}{21\eta {Q_1}^2}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5\overline{N_2Q_2}}{21\eta {Q_1}^2}(\gamma_{12})^2\\&\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{5\overline{N_2}Q_2}{21\eta {Q_1}^2}(\gamma_{12})^2(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5N_2Q_2}{21\eta {Q_1}^2}\gamma_{12}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5N_2}{21\eta Q_1}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}\\&+\frac{5\overline{N_2}}{21\eta Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5\overline{N_2}}{21\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5\overline{N_2}Q_2}{21\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5\overline{N_2Q_2}}{21\eta {Q_1}^2}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\Big|_{x_1}\\&+\frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\\&+\frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}+\left\{\frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}+\frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta {Q_1}^2}\gamma_{12}-\frac{\overline{N_2}}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}\right.\\&\left.+\frac{\overline{N_2}Q_2}{3\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\}\left.\frac{df_1(x)}{dx}\right|_{x_1}-\left\{\frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}+\frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}}-\frac{N_2Q_2}{3\eta {Q_1}^2}(\gamma_{12})^2-\frac{\overline{N_2}}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\right.\\&\left.-\frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)+\frac{\overline{N_2}Q_2}{3\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^2\right\}\left.\frac{df_2(x)}{dx}\right|_{x_2}\end{aligned}$$

$$= \frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{1\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7Q_1}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}(\gamma_{12})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}$$
$$- \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \left\{\frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\right.$$
$$\left. - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5N_2}{21\eta Q_2}(\gamma_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2Q_2}}{21\eta{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$- \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5N_2Q_2}{21\eta{Q_1}^2}(\gamma_{12})^2\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2}Q_2}{21\eta{Q_1}^2}(\gamma_{12})^2(\beta_{2\overline{2}})^{\frac{3}{2}}$$
$$\frac{df_2(x)}{dx}\Big|_{x_2} + \left\{\frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12} - \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}} + \frac{N_2Q_2}{3\eta{Q_1}^2}(\gamma_{12})^2 + \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}}\right.$$
$$\left.(\beta_{2\overline{2}})^{\frac{3}{2}} + \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}Q_2}{3\eta{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^2\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5N_2Q_2}{21\eta{Q_1}^2}\gamma_{12}\frac{df_1(x)}{dx}\Big|_{x_1}$$
$$- \frac{5N_2}{21\eta Q_1}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5\overline{N_2}}{21\eta Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5\overline{N_2}}{21\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5\overline{N_2}Q_2}{21\eta{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_1(x)}{dx}\Big|_{x_1}$$
$$+ \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\Big|_{x_1} + \left\{\frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta{Q_1}^2}\gamma_{12} - \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}\right.$$
$$\left. + \frac{\overline{N_2}Q_2}{3\eta{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\}\frac{df_1(x)}{dx}\Big|_{x_1} \tag{103}$$

At second,

$$K_{11}\cdot 2 = \int_0^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{104}$$
$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

Substituting values from appendix

$$= (\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left[\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left\{\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{d\overline{f_2(x)}}{dx}\right\} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left\{\frac{df_2(x)}{dx}\Big|_{x_2} - \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right\} + \frac{1}{Q_1}\left\{\frac{df_1(x)}{dx}\Big|_{x_1}\right.\right.$$
$$\left.\left. - \gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2}\right\}\right] + (\gamma_{12})^{\frac{3}{2}}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\left[\frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left\{\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{df_2(x)}{dx}\right\} + \frac{1}{Q_1}\left\{\frac{df_1(x)}{dx}\Big|_{x_1} - \gamma_{12}\frac{df_2(x)}{dx}\Big|_{x_2}\right\}\right]$$
$$+ \int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\left[\frac{1}{Q_1}\left\{\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{df_1(x)}{dx}\right\}\right]$$

$$= \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$- \frac{1}{Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}$$
$$- \frac{1}{Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{Q_2}(\gamma_{12})^3\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$- \frac{1}{Q_2}(\gamma_{12})^3\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_1}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{(\gamma_{12})^{\frac{5}{2}}}{Q_1}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$
$$+ \frac{1}{Q_1}\int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{1}{Q_1}\int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}$$

The second term, eighth term and last term gets cancelled with $K_{11}\cdot 1$,

$$
\begin{aligned}
&= \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{1}{Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2}\\
&- \frac{1}{Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{1}{Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Bigg|_{x_1}\\
&- \frac{1}{Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2} + \frac{1}{Q_2}(\gamma_{12})^3\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2}\\
&+ \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_1}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Bigg|_{x_1} - \frac{(\gamma_{12})^{\frac{5}{2}}}{Q_1}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2} + \frac{1}{Q_1}\int_{x_2}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Bigg|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\frac{\overline{N_2}}{3\eta}\frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{1}{Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{\overline{N_2}}{3\eta}\frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{1}{Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{\overline{N_2}}{3\eta}\frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{Q_1}\frac{\overline{N_2}}{3\eta}\frac{df_1(x)}{dx}\Bigg|_{x_1}\\
&- \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}}{Q_1}\frac{\overline{N_2}}{3\eta}\frac{df_2(x)}{dx}\Bigg|_{x_2} + \frac{(\gamma_{12})^3}{{Q_2}^2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}\frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{{Q_2}^2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}\frac{df_2(x)}{dx}\Bigg|_{x_2}\\
&+ \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_1Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}\frac{df_1(x)}{dx}\Bigg|_{x_1} - \frac{(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_1Q_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}\frac{df_1(x)}{dx}\Bigg|_{x_1}\\
&- \frac{(\gamma_{12})^{\frac{5}{2}}}{Q_1Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}\frac{df_2(x)}{dx}\Bigg|_{x_2} + \frac{(\gamma_{12})^{\frac{5}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_1Q_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}\frac{df_2(x)}{dx}\Bigg|_{x_2}\\
&+ \frac{1}{{Q_1}^2}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_1}\frac{df_1(x)}{dx}_{x_1} - \frac{\gamma_{12}}{{Q_1}^2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}\frac{df_1(x)}{dx}\Bigg|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{\overline{N_2}}{3\eta}\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta}\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}}{Q_2}\frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{\overline{N_2}}{3\eta}\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\frac{d\overline{f_2(x)}}{dx}\Bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta}\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{Q_1}\frac{df_1(x)}{dx}\Bigg|_{x_1}\\
&- \frac{\overline{N_2}}{3\eta}\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}}{Q_1}\frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{\overline{N_2Q_2}}{3\eta}\frac{(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{{Q_2}^2}\frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{\overline{N_2Q_2}}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\Bigg|_{x_1} + \frac{\overline{N_2Q_2}}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{5}{2}}\\
&\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Bigg|_{x_2} + \frac{N_2Q_2}{3\eta}\frac{(\gamma_{12})^3}{{Q_2}^2}\frac{df_2(x)}{dx}\Bigg|_{x_2} + \frac{\overline{N_2Q_2}}{3\eta{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{\overline{N_2}Q_2}{3\eta{Q_2}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\frac{df_2(x)}{dx}\Bigg|_{x_2}\\
&+ \frac{N_2Q_2}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Bigg|_{x_1} + \frac{\overline{N_2Q_2}}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\Bigg|_{x_1} - \frac{\overline{N_2}Q_2}{3\eta Q_1Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Bigg|_{x_1}\\
&- \frac{N_2Q_2}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{5}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2} - \frac{\overline{N_2Q_2}}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{5}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\Bigg|_{x_2} + \frac{\overline{N_2}Q_2}{3\eta Q_1Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{5}{2}}\frac{df_2(x)}{dx}\Bigg|_{x_2}\\
&- \frac{N_2Q_2}{3\eta{Q_1}^2}\gamma_{12}\frac{df_1(x)}{dx}\Bigg|_{x_1} - \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}\frac{df_1(x)}{dx}\Bigg|_{x_1} + \frac{\overline{N_2}Q_2}{3\eta{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_1(x)}{dx}\Bigg|_{x_1} + \frac{N_1Q_1}{3\eta{Q_1}^2}\frac{df_1(x)}{dx}\Bigg|_{x_1}\\
&+ \frac{N_2Q_2}{3\eta{Q_1}^2}\gamma_{12}\frac{df_1(x)}{dx}\Bigg|_{x_1} - \frac{N_2Q_1}{3\eta{Q_1}^2}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Bigg|_{x_1} + \frac{\overline{N_2}Q_1}{3\eta{Q_1}^2}\left\{(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - (\beta_{1\overline{2}})^{\frac{3}{2}}\right\}\frac{df_1(x)}{dx}\Bigg|_{x_1}\\
&- \frac{\overline{N_2}Q_2}{3\eta{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_1(x)}{dx}\Bigg|_{x_1} + \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\Bigg|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{\overline{N_2}}{3\eta}\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_2(x)}{dx}\bigg|_{x_2} \\
&- \frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{\overline{N_2Q_2}}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{5}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{N_2}{3\eta Q_2}(\gamma_{12})^3\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^3 \\
&\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}}\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{\overline{N_2Q_2}}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{5}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2} \\
&+ \frac{\overline{N_2}}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{5}{2}}\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\overline{N_2Q_2}}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \frac{\overline{N_2Q_2}}{3\eta Q_1Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\overline{N_2}}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{N_2Q_2}{3\eta {Q_1}^2}\gamma_{12}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \frac{\overline{N_2}Q_2}{3\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{N_1}{3\eta Q_1}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{N_2Q_2}{3\eta {Q_1}^2}\gamma_{12}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \frac{\overline{N_2}}{3\eta Q_1}\left\{(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - (\beta_{1\overline{2}})^{\frac{3}{2}}\right\}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\overline{N_2}Q_2}{3\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&= \frac{\overline{N_2}}{3\eta}\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_2(x)}{dx}\bigg|_{x_2} \\
&+ \frac{N_2}{3\eta Q_2}(\gamma_{12})^3\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}}\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{\overline{N_2}}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{5}{2}}\frac{df_2(x)}{dx}\bigg|_{x_2} \\
&+ \frac{N_1}{3\eta Q_1}\frac{df_1(x)}{dx}\bigg|_{x_1}
\end{aligned}
\tag{105}
$$

Therefore, $K_{11} = K_{11}\cdot 1 + K_{11}\cdot 2$

$$
\begin{aligned}
&= \frac{4}{7Q_2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\bigg|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\beta_{1\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\bigg|_{\overline{x_2}} + \frac{4}{7Q_1}(\gamma_{12})^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\bigg|_{x_2} - \frac{4}{7Q_2}(\gamma_{12})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\bigg|_{x_2} \\
&- \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\bigg|_{x_1} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\beta_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta {Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^3\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \left\{\frac{\overline{N_2}}{3\eta}\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\right. \\
&\left. - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) + \frac{\overline{N_2}}{3\eta}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2\frac{\overline{Q_2}}{{Q_2}^2} - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right\}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \\
&+ \frac{5N_2}{21\eta Q_2}(\gamma_{12})^3\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{5\overline{N_2Q_2}}{21\eta {Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{5\overline{N_2}}{21\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{5N_2Q_2}{21\eta {Q_1}^2}(\gamma_{12})^2\frac{df_2(x)}{dx}\bigg|_{x_2} \\
&- \frac{5\overline{N_2Q_2}}{21\eta {Q_1}^2}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{5\overline{N_2}}{21\eta}\frac{Q_2}{{Q_1}^2}(\gamma_{12})^2(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\bigg|_{x_2} + \left\{\frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}} - \frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}\right. \\
&+ \frac{N_2}{3\eta Q_2}(\gamma_{12})^3 - \frac{\overline{N_2}}{3\eta Q_2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3 - \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}} + \frac{\overline{N_2}}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{5}{2}} + \frac{\overline{N_2}}{3\eta Q_2}(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta {Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) \\
&- \frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12} - \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}} + \frac{N_2}{3\eta}\frac{Q_2}{{Q_1}^2}(\gamma_{12})^2 + \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{12})^{\frac{5}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} + \frac{\overline{N_2}}{3\eta}\frac{\overline{Q_2}}{{Q_1}^2}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) \\
&\left. - \frac{\overline{N_2}Q_2}{3\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^2\right\}\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{5N_2Q_2}{21\eta {Q_1}^2}\gamma_{12}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{5N_2}{21\eta Q_1}(\gamma_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \frac{5\overline{N_2}}{21\eta Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{5\overline{N_2}}{21\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{5\overline{N_2}Q_2}{21\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{5\overline{N_2Q_2}}{21\eta {Q_1}^2}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \left\{\frac{N_1}{3\eta Q_1} + \frac{\overline{N_2}}{3\eta Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta {Q_1}^2}\gamma_{12} - \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12}\right. \\
&\left. + \frac{\overline{N_2}Q_2}{3\eta {Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\}\frac{df_1(x)}{dx}\bigg|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
K_{11} = \frac{4}{7}\left[\frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^{\frac{5}{2}}(\gamma_{12})^3}{Q_2} - \frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\right]\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7}\left[\frac{(\gamma_{12})^{\frac{5}{2}}}{Q_1} - \frac{(\gamma_{12})^3}{Q_2}\right]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} \\
+ \left[\frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} + \frac{\overline{Q_2}}{{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2 - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\right\} + \frac{5\overline{N_2}}{21\eta}\left\{\frac{(\beta_{1\overline{2}})^3}{\overline{Q_2}}\right. \\
\left. - \frac{\overline{Q_2}}{{Q_2}^2}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)^2(\gamma_{12})^3\right\}\left]\frac{d\overline{f_2(x)}}{dx}\right|_{\overline{x_2}} + \left[\frac{\overline{N_2}}{3\eta}\left\{\frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}}{Q_2} - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}}{Q_1} - \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3}{Q_2} + \frac{(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{5}{2}}}{Q_1}\right. \right. \\
+ \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}(\gamma_{12})^{\frac{3}{2}}}{Q_2} - \frac{\overline{Q_2}}{{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}\gamma_{12}}{Q_1} + \frac{(\gamma_{12})^{\frac{5}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}}}{Q_1} + \frac{\overline{Q_2}}{{Q_1}^2}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{Q_2}{{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^2\Big\} \\
+ \frac{N_2}{3\eta}\left\{\frac{(\gamma_{12})^3}{Q_2} - \frac{2(\gamma_{12})^{\frac{5}{2}}}{Q_1} + \frac{Q_2}{{Q_1}^2}(\gamma_{12})^2\right\} + \frac{5}{21\eta}\left\{\frac{N_2}{Q_2}(\gamma_{12})^3 + \frac{\overline{N_2Q_2}}{{Q_2}^2}(\gamma_{12})^3\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}}{\overline{Q_2}}(\beta_{2\overline{2}})^{\frac{3}{2}}(\gamma_{12})^3\right. \\
\left. - \frac{N_2Q_2}{{Q_1}^2}(\gamma_{12})^2 - \frac{\overline{N_2Q_2}}{{Q_1}^2}(\gamma_{12})^2\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) + \frac{\overline{N_2}Q_2}{{Q_1}^2}(\gamma_{12})^2(\beta_{2\overline{2}})^{\frac{3}{2}}\right\}\left]\frac{df_2(x)}{dx}\right|_{x_2} + \left[\frac{N_1}{3\eta Q_1} - \frac{N_2Q_2}{3\eta{Q_1}^2}\gamma_{12} + \frac{N_2}{3\eta Q_1}(\gamma_{12})^{\frac{3}{2}}\right. \\
+ \frac{\overline{N_2}}{3\eta Q_1}\left\{(\beta_{1\overline{2}})^{\frac{3}{2}} - (\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{Q_2}}{Q_1}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12} + \frac{Q_2}{Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12}\right\} + \frac{5}{21\eta}\left\{\frac{N_1}{Q_1} + \frac{N_2Q_2}{{Q_1}^2}\gamma_{12} - \frac{N_2}{Q_1}(\gamma_{12})^{\frac{3}{2}}\right. \\
\left. \left. + \frac{\overline{N_2}}{Q_1}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2}}{Q_1}(\beta_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2}Q_2}{{Q_1}^2}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12} + \frac{\overline{N_2Q_2}}{{Q_1}^2}\gamma_{12}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\right\}\right]\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
\tag{106}
$$

Appendix for K_{11}

1.

$$\int_0^x dx\sqrt{x}(f_1(x))^{\frac{3}{2}}$$

In region $0 < x \leq \overline{x_2}$, $\overline{f_2(x)}$ is dominant

$$
\begin{aligned}
&= (\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^x dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \\
&= (\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^x dx\frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2} \\
&= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_0^x dxx\frac{d^2\overline{f_2(x)}}{dx^2} \\
&= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]
\end{aligned}
\tag{107}
$$

In region $\overline{x_2} < x \leq x_2$, $f_2(x)$ is dominant

$$
\begin{aligned}
&= \int_0^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} + \int_{\overline{x_2}}^x dx\sqrt{x}(f_1(x))^{\frac{3}{2}} \\
&= (\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} + (\gamma_{12})^{\frac{3}{2}}\int_{\overline{x_2}}^x dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \\
&= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^x dxx\frac{d^2f_2(x)}{dx^2} \\
&= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}^x \\
&= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right] - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}} \\
&= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right] - \frac{(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}
\end{aligned}
$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right] \tag{108}$$

In region $x_2 < x \leq x_1$, $f_1(x)$ is dominant

$$= \int_0^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} + \int_{x_2}^{x} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}$$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}}\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} + (\gamma_{12})^{\frac{3}{2}}\int_{\overline{x_2}}^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} + \int_{x_2}^{x} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^{x_2} dxx\frac{d^2f_2(x)}{dx^2} + \frac{1}{Q_1}\int_{x_2}^{x} dxx\frac{d^2f_1(x)}{dx^2}$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}}^{x_2} + \frac{1}{Q_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2}^{x}$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2} - \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{\overline{x_2}} + \frac{1}{Q_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right] - \frac{1}{Q_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_2}$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left\{\frac{N_2Q_2}{3\eta} + \frac{\overline{N_2Q_2}}{3\eta}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}Q_2}{3\eta}(\beta_{2\overline{2}})^{\frac{3}{2}}\right\} + \frac{1}{Q_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]$$
$$- \frac{\gamma_{12}}{Q_1}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) + \frac{N_2}{3\eta}(\gamma_{12})^{\frac{3}{2}} + \frac{\overline{N_2Q_2}}{3\eta Q_2}(\gamma_{12})^{\frac{3}{2}}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right) - \frac{\overline{N_2}}{3\eta}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta Q_1}\gamma_{12}$$
$$- \frac{\overline{N_2Q_2}}{3\eta Q_1}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12} + \frac{\overline{N_2}Q_2}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12} + \frac{1}{Q_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]$$

$$= \frac{\overline{N_2}}{3\eta}(\beta_{1\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta}(\gamma_{12})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta Q_1}\gamma_{12} - \frac{\overline{N_2}}{3\eta}(\gamma_{12})^{\frac{3}{2}}(\beta_{2\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta Q_1}\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\gamma_{12} + \frac{\overline{N_2}Q_2}{3\eta Q_1}(\beta_{2\overline{2}})^{\frac{3}{2}}\gamma_{12} + \frac{1}{Q_1}\left[x\frac{df_1(x)}{dx} - f_1(x)\right] \tag{109}$$

2.

$$\int_x^{x_1} dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

In region $0 < x \leq \overline{x_2}$,

$$= \int_x^{\overline{x_2}} dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + \int_{\overline{x_2}}^{x_2} dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + \int_{x_2}^{x_1} dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$= (\beta_{1\overline{2}})^{\frac{3}{2}}\int_x^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + (\gamma_{12})^{\frac{3}{2}}\int_{\overline{x_2}}^{x_2} dx\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \int_{x_2}^{x_1} dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_x^{\overline{x_2}} dx\frac{d^2\overline{f_2(x)}}{dx^2} + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_{\overline{x_2}}^{x_2} dx\frac{d^2f_2(x)}{dx^2} + \frac{1}{Q_1}\int_{x_2}^{x_1} dx\frac{d^2f_1(x)}{dx^2}$$

$$= \frac{(\beta_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left[\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{d\overline{f_2(x)}}{dx}\right] + \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[\frac{df_2(x)}{dx}\bigg|_{x_2} - \left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}\right] + \frac{1}{Q_1}\left[\frac{df_1(x)}{dx}\bigg|_{x_1} - \gamma_{12}\frac{df_2(x)}{dx}\bigg|_{x_2}\right] \tag{110}$$

In region $\overline{x_2} < x \leq x_2$,

$$= \int_x^{x_2} dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + \int_{x_2}^{x_1} dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$= (\gamma_{12})^{\frac{3}{2}}\int_x^{x_2} dx\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \int_{x_2}^{x_1} dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$= \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\int_x^{x_2} dx\frac{d^2f_2(x)}{dx^2} + \frac{1}{Q_1}\int_{x_2}^{x_1} dx\frac{d^2f_1(x)}{dx^2}$$

$$= \frac{(\gamma_{12})^{\frac{3}{2}}}{Q_2}\left[\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{df_2(x)}{dx}\right] + \frac{1}{Q_1}\left[\frac{df_1(x)}{dx}\bigg|_{x_1} - \gamma_{12}\frac{df_2(x)}{dx}\bigg|_{x_2}\right] \tag{111}$$

In region $x_2 < x \leq x_1$,

$$\begin{aligned}
&\int_x^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \\
&= \frac{1}{Q_1} \int_x^{x_1} dx \frac{d^2 f_1(x)}{dx^2} \\
&= \frac{1}{Q_1} \left[\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{df_1(x)}{dx} \right]
\end{aligned} \tag{112}$$

3. From Normalization condition,

$$\int_0^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta} \tag{113}$$

4.

$$\begin{aligned}
&\int_{\overline{x_2}}^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \\
&= \int_{\overline{x_2}}^{x_2} dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2} \\
&= \frac{1}{Q_2} \int_{\overline{x_2}}^{x_2} dx x \frac{d^2 f_2(x)}{dx^2} \\
&= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{\overline{x_2}}^{x_2} \\
&= \frac{1}{Q_2} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} - \frac{\left(\frac{\beta_{1\overline{2}}}{\gamma_{12}}\right)}{Q_2} \left[x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{\overline{x_2}}
\end{aligned} \tag{114}$$

5.

$$\begin{aligned}
&\int_{x_2}^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \\
&= \int_{x_2}^{x_1} dx \frac{x}{Q_1} \frac{d^2 f_1(x)}{dx^2} \\
&= \frac{1}{Q_1} \int_{x_2}^{x_1} dx x \frac{d^2 f_1(x)}{dx^2} \\
&= \frac{1}{Q_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_2}^{x_1} \\
&= \frac{1}{Q_1} \left[x \frac{df_1(x)}{dx} - f_1(x) \right]_{x_1} - \frac{\gamma_{12}}{Q_1} \left[x \frac{df_2(x)}{dx} - f_2(x) \right]_{x_2}.
\end{aligned} \tag{115}$$

Case II

$0 < x \leq x_2$, all three functions are present but anti-charm quark is dominant

$x_2 < x \leq \overline{x_2}$, only charm quark and light quarks are present and charm quark is dominant

$\overline{x_2} < x \leq x_1$, only light quark is present.

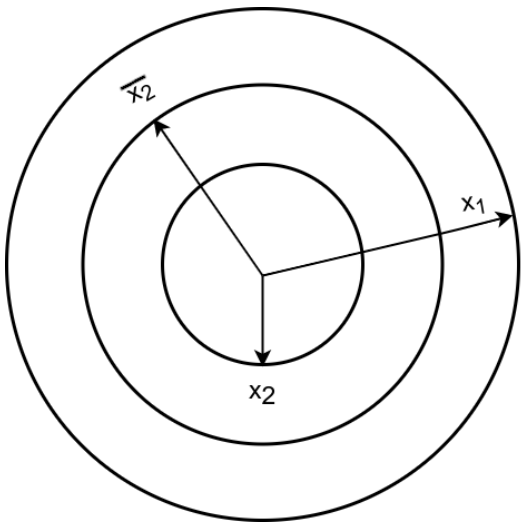


Figure 2: Distribution of quarks for the case of $\eta = 1$

- **In region** $0 < x < x_2$

In this region all three functions are present but we use the linear relationship of other functions with $f_2(x)$ as

$$f_1(x) = \beta_{12} f_2(x)$$
$$\overline{f_2(x)} = \beta_{\overline{2}2} f_2(x)$$

where, β_{12} & $\beta_{\overline{2}2}$ are proportionality constants

Equation 3 gives

$$\overline{\alpha_2}\frac{d^2\overline{f_2(x)}}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right\}$$
$$\beta_{\overline{2}2}\frac{d^2 f_2(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right\}$$
$$\frac{d^2 f_2(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)\overline{\alpha_2}\beta_{\overline{2}2}}\left\{g_1(\beta_{12})^{\frac{3}{2}} + g_2\right\}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{116}$$

Equation 4 gives

$$\alpha_1\frac{d^2 f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1 g_1 - 1)}{N_1}\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

We have, $\alpha_1 = \frac{m^1}{m^1} = 1$. So that,

$$\beta_{12}\frac{d^2 f_2(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1 g_1 - 1)}{N_1}(\beta_{12})^{\frac{3}{2}}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$
$$\frac{d^2 f_2(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)\beta_{12}}\left[\frac{(N_1 g_1 - 1)}{N_1}(\beta_{12})^{\frac{3}{2}} + g_2\right]\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{117}$$

Equation 5 gives

$$\alpha_2\frac{d^2 f_2(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\overline{g_2}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)}\left[\frac{(N_2 g_2 - 1)}{N_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$
$$\alpha_2\frac{d^2 f_2(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\overline{g_2}(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)}\left[\frac{(N_2 g_2 - 1)}{N_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_1(\beta_{12})^{\frac{3}{2}}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$
$$\frac{d^2 f_2(x)}{dx^2} = \left[-\frac{6\eta}{5(5\eta-1)\alpha_2}\overline{g_2}(\beta_{\overline{2}2})^{\frac{3}{2}} - \frac{9\eta}{10(5\eta-1)\alpha_2}\left\{\frac{(N_2 g_2 - 1)}{N_2} + g_1(\beta_{12})^{\frac{3}{2}}\right\}\right]\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{118}$$

In a single form eq^{n}s 116, 117, 118 can be written as

$$\frac{d^2 f_2(x)}{dx^2} = Q_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$

where,

$$Q_2 = \begin{cases} -\dfrac{6\eta}{5(5\eta-1)\overline{\alpha_2}\beta_{\overline{2}2}}\left\{g_1(\beta_{12})^{\frac{3}{2}} + g_2\right\} \\ -\dfrac{9\eta}{10(5\eta-1)\beta_{12}}\left[\dfrac{(N_1 g_1 - 1)}{N_1}(\beta_{12})^{\frac{3}{2}} + g_2\right] \\ -\dfrac{6\eta}{5(5\eta-1)\alpha_2}\overline{g_2}(\beta_{\overline{2}2})^{\frac{3}{2}} - \dfrac{9\eta}{10(5\eta-1)\alpha_2}\left\{\dfrac{(N_2 g_2 - 1)}{N_2} + g_1(\beta_{12})^{\frac{3}{2}}\right\} \end{cases} \tag{119}$$

- **In region** $x_2 < x < \overline{x_2}$

In this region, two functions $\overline{f_2(x)}$ & $f_1(x)$ are present and we use linear relationship of $f_1(x)$ with $\overline{f_2(x)}$ as

$$f_1(x) = \gamma_{1\bar{2}}\overline{f_2(x)}$$

Equation 3 gives

$$\begin{aligned}
\overline{\alpha_2}\frac{d^2\overline{f_2(x)}}{dx^2} &= -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right\} \\
\overline{\alpha_2}\frac{d^2\overline{f_2(x)}}{dx^2} &= -\frac{6\eta}{5(5\eta-1)}g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \\
\frac{d^2\overline{f_2(x)}}{dx^2} &= -\frac{6\eta}{5(5\eta-1)\overline{\alpha_2}}g_1(\gamma_{1\bar{2}})^{\frac{3}{2}}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}
\end{aligned} \tag{120}$$

sdsfEquation 4 gives

$$\alpha_1\frac{d^2f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)}\left[\frac{(N_1g_1-1)}{N_1}\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} + g_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

We have, $\alpha_1 = \frac{m^1}{m^1} = 1$. So that,

$$\begin{aligned}
\frac{d^2f_1(x)}{dx^2} &= -\frac{9\eta}{10(5\eta-1)}\frac{(N_1g_1-1)}{N_1}\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \\
\gamma_{1\bar{2}}\frac{d^2\overline{f_2(x)}}{dx^2} &= -\frac{9\eta}{10(5\eta-1)}\frac{(N_1g_1-1)}{N_1}(\gamma_{1\bar{2}})^{\frac{3}{2}}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \\
\frac{d^2\overline{f_2(x)}}{dx^2} &= -\frac{9\eta}{10(5\eta-1)}\frac{(N_1g_1-1)}{N_1}(\gamma_{1\bar{2}})^{\frac{1}{2}}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}
\end{aligned} \tag{121}$$

Equation 5 gives

$$\alpha_2\frac{d^2f_2(x)}{dx^2} = -\frac{6\eta}{5(5\eta-1)}\overline{g_2}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)}\left[\frac{(N_2g_2-1)}{N_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right]$$

$$\begin{aligned}
0 &= -\frac{6\eta}{5(5\eta-1)}\overline{g_2}\frac{(\overline{f_2(x)}}{\sqrt{x}} - \frac{9\eta}{10(5\eta-1)}g_1(\gamma_{1\bar{2}})^{\frac{3}{2}}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \\
0 &= \left[-\frac{6\eta}{5(5\eta-1)}\overline{g_2} - \frac{9\eta}{10(5\eta-1)}g_1(\gamma_{1\bar{2}})^{\frac{3}{2}}\right]\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \\
0 &= \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}
\end{aligned}$$

This leads to meaningless result. So we drop this equation.

In a single form eq^{n}s 120, 121 can be written as

$$\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{Q_2}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

where,

$$\begin{aligned}
\overline{Q_2} &= -\frac{6\eta}{5(5\eta-1)\overline{\alpha_2}}g_1(\gamma_{1\bar{2}})^{\frac{3}{2}} \\
&= -\frac{9\eta}{10(5\eta-1)}\frac{(N_1g_1-1)}{N_1}(\gamma_{1\bar{2}})^{\frac{1}{2}}
\end{aligned} \tag{122}$$

- **In region** $\overline{x_2} < x \leq x_1$

In this region, $f_2(x) = 0$ & $\overline{f_2(x)} = 0$.

Equation 3 gives

$$0 = -\frac{6\eta}{5(5\eta-1)}\left\{g_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}\right\} \tag{123}$$

Equation 4 gives

$$\alpha_1 \frac{d^2 f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)} \left[\frac{(N_1 g_1 - 1)}{N_1} \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \right] \tag{124}$$

Equation 5 gives

$$0 = -\frac{9\eta}{10(5\eta-1)} \left[g_1 \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \right] \tag{125}$$

Equations (123), (125) results $f_1(x) = 0$, which leads to meaningless result. So we drop these two equations and consider only equation (124).

$$\alpha_1 \frac{d^2 f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)} \left[\frac{(N_1 g_1 - 1)}{N_1} \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \right]$$

We have, $\alpha_1 = \frac{m^1}{m^1} = 1$. So that,

$$\frac{d^2 f_1(x)}{dx^2} = -\frac{9\eta}{10(5\eta-1)} \left[\frac{(N_1 g_1 - 1)}{N_1} \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \right]$$

$$\frac{d^2 f_1(x)}{dx^2} = Q_1 \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{126}$$

$$\text{where, } Q_1 = \frac{-9\eta}{10(5\eta-1)} \frac{(N_1 g_1 - 1)}{N_1} \tag{127}$$

1 Preparing the appendix

1.1 Appendix A

$$f'' = Q \frac{f^{\frac{3}{2}}}{\sqrt{x}} \tag{i}$$

$$f'' f = Q \frac{f^{\frac{5}{2}}}{\sqrt{x}}$$

$$\int_0^{x_m} \frac{f f''}{Q} = \int_0^{x_m} \frac{f^{\frac{5}{2}}}{\sqrt{x}} dx \tag{ii}$$

Integrating by parts in L.H.S of eq^n[ii], we get

$$\longrightarrow \int_0^{x_m} \frac{(f(x))^{\frac{5}{2}}}{\sqrt{x}} dx = \frac{f f' - \int_0^{x_{max}} f'^2 dx}{Q} \tag{iii}$$

Multiplying eq^n [i] by xf' on both sides, we get

$$xf' f'' = Q\sqrt{x} f' f^{\frac{3}{2}}$$

$$x \cdot \frac{1}{2} \frac{d(f')^2}{dx} = Q\sqrt{x} f' f^{\frac{3}{2}}$$

$$\frac{x}{2} \frac{d(f')^2}{dx} = Q\sqrt{x} \frac{df^{\frac{5}{2}}}{dx} \cdot \frac{2}{5} \tag{iv}$$

$$x \frac{d(f')^2}{dx} = \frac{4}{5} Q\sqrt{x} \frac{df^{\frac{5}{2}}}{dx}$$

Integrating,

$$\int x \frac{d(f')^2}{dx} dx = \frac{4}{5} Q \int dx \sqrt{x} \frac{df^{\frac{5}{2}}}{dx}$$

$$xf'^2 - \int_0^{x_{max}} f'^2 dx = \frac{4}{5} Q \left[\sqrt{x} f^{\frac{5}{2}} - \int_0^{x_m} \frac{1}{2\sqrt{x}} f^{\frac{5}{2}} dx \right]$$

$$\int_0^{x_m} f'^2 dx = xf' - \frac{4}{5} Q \left[\sqrt{x} f^{\frac{5}{2}} - \int_0^{x_m} \frac{1}{2\sqrt{x}} f^{\frac{5}{2}} dx \right]$$

Using in eq^n[iii],

$$\int_0^{x_m} \frac{(f(x))^{\frac{5}{2}}}{\sqrt{x}}dx = \frac{1}{Q}\left[ff' - xf'^2 + \frac{4}{5}Q\left\{\sqrt{x}f^{\frac{5}{2}} - \int_0^{x_m} \frac{1}{2\sqrt{x}}f^{\frac{5}{2}}dx\right\}\right]$$

$$\int_0^{x_m} \frac{(f(x))^{\frac{5}{2}}}{\sqrt{x}}dx = \frac{ff'}{Q} - \frac{xf'^2}{Q} + \frac{4}{5}\sqrt{x}f^{\frac{5}{2}} - \frac{2}{5}\int_0^{x_m} \frac{f^{\frac{5}{2}}}{\sqrt{x}}dx$$

$$\frac{7}{5}\int_0^{x_m} \frac{(f(x))^{\frac{5}{2}}}{\sqrt{x}}dx = \frac{f'(f - xf')}{Q} + \frac{4}{5}\sqrt{x}f^{\frac{5}{2}}$$

$$\boxed{\longrightarrow \frac{(f(x))^{\frac{5}{2}}}{\sqrt{x}}dx = \frac{4}{7}\sqrt{x}f^{\frac{5}{2}} - \frac{5}{7Q}f'(xf' - f)} \tag{vi}$$

So that,

$$\int_0^{x_2} \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}dx = \frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5}{7Q_2}\frac{df_2(x)}{dx}\left(x\frac{df_2(x)}{dx} - f_2(x)\right)\Big|_{x_2} \tag{vii}$$

$$\int_{x_2}^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}dx = \frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{x_2} - \frac{5}{7\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\left(x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right)\Big|_{\overline{x_2}} + \frac{5}{7Q_2}\frac{d\overline{f_2(x)}}{dx}\left(x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right)\Big|_{x_2} \tag{viii}$$

$$\int_{\overline{x_2}}^{x_1} \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}dx = \frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} - \frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5}{7Q_1}\frac{df_1(x)}{dx}\left(x\frac{df_1(x)}{dx} - f_1(x)\right)\Big|_{x_1} + \frac{5}{7\overline{Q_2}}\frac{df_1(x)}{dx}\left(x\frac{df_1(x)}{dx} - f_1(x)\right)\Big|_{\overline{x_2}} \tag{ix}$$

1.2 Appendix B

In region $0 < x < x_2$

$$\frac{d^2f_2(x)}{dx^2} = Q_2\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$\frac{x}{Q_2}\frac{d^2f_2(x)}{dx^2} = \sqrt{x}(f_2(x))^{\frac{3}{2}}$$

$$\int_0^{x_2} dx\frac{x}{Q_2}\frac{d^2f_2(x)}{dx^2} = \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$

Since,

$$\boxed{\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} = \frac{N_2}{3\eta}}$$

So that,

$$\int_0^{x_2} dx\frac{x}{Q_2}\frac{d^2f_2(x)}{dx^2} = \frac{N_2}{3\eta}$$

$$\int_0^{x_2} dxx\frac{d^2f_2(x)}{dx^2} = \frac{N_2Q_2}{3\eta}$$

$$x\frac{df_2(x)}{dx} - \int_0^{x_2} \frac{df_2(x)}{dx}dx = \frac{N_2Q_2}{3\eta}$$

$$\left(x\frac{df_2(x)}{dx} - f_2(x)\right)\Big|_{x_2} = \frac{N_2Q_2}{3\eta} \tag{x}$$

In region $x_2 < x < \overline{x_2}$

$$\frac{d^2\overline{f_2(x)}}{dx^2} = \overline{Q_2}\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

$$\frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2} = \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

$$\int_{x_2}^{\overline{x_2}} dx \frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2} = \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

Since,

$$\boxed{\int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} = \int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} + \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} = \frac{\overline{N_2}}{3\eta}}$$

So that,

$$\int_{x_2}^{\overline{x_2}} dx \frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2} = \frac{\overline{N_2}}{3\eta} - \underbrace{\int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}}_{\overline{f_2(x)}=\beta_{\overline{2}\overline{2}}f_2(x)}$$

$$\int_{x_2}^{\overline{x_2}} dx \frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2} = \frac{\overline{N_2}}{3\eta} - (\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$

$$\int_{x_2}^{\overline{x_2}} dx \frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2} = \frac{\overline{N_2}}{3\eta} - (\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2}{3\eta}$$

$$\int_{x_2}^{\overline{x_2}} dx x\frac{d^2\overline{f_2(x)}}{dx^2} = \frac{\overline{N_2Q_2}}{3\eta} - (\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2\overline{Q_2}}{3\eta}$$

$$\left(x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right)\Bigg|_{x_2}^{\overline{x_2}} = \frac{\overline{N_2Q_2}}{3\eta} - (\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2\overline{Q_2}}{3\eta}$$

$$\left(x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right)\Bigg|_{\overline{x_2}} = \frac{\overline{N_2Q_2}}{3\eta} - (\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2\overline{Q_2}}{3\eta} + \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{N_2Q_2}{3\eta}$$

In region $\overline{x_2} < x < x_1$

$$\frac{d^2f_1(x)}{dx^2} = Q_1\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

$$\frac{x}{Q_1}\frac{d^2f_1(x)}{dx^2} = \sqrt{x}(f_1(x))^{\frac{3}{2}}$$

$$\int_{\overline{x_2}}^{x_1} dx \frac{x}{Q_1}\frac{d^2f_1(x)}{dx^2} = \int_{\overline{x_2}}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}$$

Since,

$$\boxed{\int_0^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \int_0^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} + \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} = \frac{N_1}{3\eta}}$$

So that,

$$\int_{\overline{x_2}}^{x_1} dx \frac{x}{Q_1}\frac{d^2f_1(x)}{dx^2} = \frac{N_1}{3\eta} - \underbrace{\int_0^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}}_{f_1(x)=\beta_{12}f_2(x)} - \underbrace{\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}}_{f_1(x)=\gamma_{1\overline{2}}\overline{f_2(x)}}$$

$$\int_{\overline{x_2}}^{x_1} dx \frac{x}{Q_1}\frac{d^2f_1(x)}{dx^2} = \frac{N_1}{3\eta} - (\beta_{12})^{\frac{3}{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} - (\gamma_{1\overline{2}})^{\frac{3}{2}}\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

$$\int_{\overline{x_2}}^{x_1} dx \frac{x}{Q_1}\frac{d^2f_1(x)}{dx^2} = \frac{N_1}{3\eta} - (\beta_{12})^{\frac{3}{2}}\frac{N_2}{3\eta} - (\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}}{3\eta} + (\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2}{3\eta}$$

$$\int_{\overline{x_2}}^{x_1} dx x\frac{d^2f_1(x)}{dx^2} = \frac{N_1Q_1}{3\eta} - (\beta_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta} - (\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta} + (\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}$$

$$\left(x\frac{df_1(x)}{dx} - f_1(x)\right)\Bigg|_{\overline{x_2}}^{x_1} = \frac{N_1Q_1}{3\eta} - (\beta_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta} - (\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta} + (\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}$$

$$\left(x\frac{df_1(x)}{dx} - f_1(x)\right)\Bigg|_{x_1} - \left(x\frac{df_1(x)}{dx} - f_1(x)\right)\Bigg|_{\overline{x_2}} = \frac{N_1Q_1}{3\eta} - (\beta_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta} - (\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta} + (\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}$$

$$\left(x\frac{df_1(x)}{dx}-f_1(x)\right)\Big|_{x_1}=\frac{N_1Q_1}{3\eta}-(\beta_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}-(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}+(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+\left(x\frac{df_1(x)}{dx}-f_1(x)\right)\Big|_{\overline{x_2}}$$

$$\left(x\frac{df_1(x)}{dx}-f_1(x)\right)\Big|_{x_1}=\frac{N_1Q_1}{3\eta}-(\beta_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}-(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}+(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+\gamma_{1\overline{2}}\left(x\frac{d\overline{f_2(x)}}{dx}-\overline{f_2(x)}\right)\Big|_{\overline{x_2}}$$

$$\left(x\frac{df_1(x)}{dx}-f_1(x)\right)\Big|_{x_1}=\frac{N_1Q_1}{3\eta}-(\beta_{12})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}-(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}+(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{N_2Q_1}{3\eta}+\gamma_{1\overline{2}}\left\{\frac{\overline{N_2Q_2}}{3\eta}-(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{N_2\overline{Q_2}}{3\eta}+\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{N_2Q_2}{3\eta}\right\}$$

$$\left(x\frac{df_1(x)}{dx}-f_1(x)\right)\Big|_{x_1}=\frac{N_1Q_1}{3\eta}+\left\{(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}-(\beta_{12})^{\frac{3}{2}}\right\}\frac{N_2Q_1}{3\eta}-(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}+\gamma_{1\overline{2}}\frac{\overline{N_2Q_2}}{3\eta}-(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{N_2\overline{Q_2}}{3\eta}+\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}\frac{N_2Q_2}{3\eta}$$

Now,
Eq^n (vii) becomes

$$\begin{aligned}\int_0^{x_2}\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}dx&=\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5}{7Q_2}\frac{df_2(x)}{dx}\left(x\frac{df_2(x)}{dx}-f_2(x)\right)\Big|_{x_2}\\&=\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5}{7Q_2}\frac{N_2Q_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2}\end{aligned}$$

Eq^n (viii) becomes

$$\begin{aligned}\int_{x_2}^{\overline{x_2}}\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}dx&=\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{x_2}-\frac{5}{7\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\left(x\frac{d\overline{f_2(x)}}{dx}-\overline{f_2(x)}\right)\Big|_{\overline{x_2}}\\&\quad+\frac{5}{7\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\left(x\frac{d\overline{f_2(x)}}{dx}-\overline{f_2(x)}\right)\Big|_{x_2}\\&=\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{4}{7}\sqrt{x}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5}{7\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\left(x\frac{d\overline{f_2(x)}}{dx}-\overline{f_2(x)}\right)\Big|_{\overline{x_2}}\\&\quad+\frac{5}{7\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\left(x\frac{df_2(x)}{dx}-f_2(x)\right)\Big|_{x_2}\\&=\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{4}{7}\sqrt{x}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5}{7\overline{Q_2}}\left\{\frac{\overline{N_2Q_2}}{3\eta}-(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{N_2\overline{Q_2}}{3\eta}+\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{N_2Q_2}{3\eta}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\\&\quad+\frac{5}{7\overline{Q_2}}\frac{N_2Q_2}{3\eta}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2}\\&=\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{4}{7}\sqrt{x}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{5}{21\eta}\left\{\overline{N_2}-(\beta_{\overline{2}2})^{\frac{3}{2}}N_2+\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{N_2Q_2}{\overline{Q_2}}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\\&\quad+\frac{5}{21\eta}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{N_2Q_2}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{x_2}\end{aligned}$$

Eq^n (ix) becomes

$$\int_{\overline{x_2}}^{x_1}\frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}dx=\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}-\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{5}{7Q_1}\frac{df_1(x)}{dx}\left(x\frac{df_1(x)}{dx}-f_1(x)\right)\Big|_{x_1}$$
$$+\frac{5}{7Q_1}\frac{df_1(x)}{dx}\left(x\frac{df_1(x)}{dx}-f_1(x)\right)\Big|_{\overline{x_2}}$$
$$=\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}-\frac{4}{7}\sqrt{x}(\gamma_{1\overline{2}})^{\frac{5}{2}}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{5}{7Q_1}\frac{df_1(x)}{dx}\left(x\frac{df_1(x)}{dx}-f_1(x)\right)\Big|_{x_1}$$
$$+\frac{5}{7Q_1}(\gamma_{1\overline{2}})^2\frac{d\overline{f_2(x)}}{dx}\left(x\frac{d\overline{f_2(x)}}{dx}-\overline{f_2(x)}\right)\Big|_{\overline{x_2}}$$
$$=\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}-\frac{4}{7}(\gamma_{1\overline{2}})^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{5}{7Q_1}\left\{\frac{N_1Q_1}{3\eta}+\left\{(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}-(\beta_{12})^{\frac{3}{2}}\right\}\frac{N_2Q_1}{3\eta}\right.$$
$$-(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{\overline{N_2}Q_1}{3\eta}+\gamma_{1\overline{2}}\frac{\overline{N_2Q_2}}{3\eta}-(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{N_2\overline{Q_2}}{3\eta}+\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}\frac{N_2Q_2}{3\eta}\left.\right\}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5}{7Q_1}(\gamma_{1\overline{2}})^2\left\{\frac{\overline{N_2Q_2}}{3\eta}\right.$$
$$\left.-(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{N_2\overline{Q_2}}{3\eta}+\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{N_2Q_2}{3\eta}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

2 Expression for Kinetic Energy

$$T=T_1+T_2 \tag{K1}$$

where,

$$T_1=\sum_I\int^{r_{max}}d^3r\frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3m^Ig_I^{\frac{2}{3}}}(n^I(r))^{\frac{5}{3}}$$
$$T_2=\sum_I\int^{r_{max}}d^3r\frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3\overline{m}^I(\overline{g_I})^{\frac{2}{3}}}(\overline{n^I(r)})^{\frac{5}{3}}$$

Since,

$$n^I(r)=\left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{3}{2}}\left(\frac{f_I(r)}{r}\right)^{\frac{3}{2}}\frac{g_I}{6\pi^2} \tag{K2}$$
$$\overline{n^I(r)}=\left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{3}{2}}\left(\frac{\overline{f_I(r)}}{r}\right)^{\frac{3}{2}}\frac{\overline{g_I}}{6\pi^2}$$

T_1 becomes

$$T_1=\frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3}\sum_I\int^{r_{max}}d^3r\frac{(n^I(r))^{\frac{5}{3}}}{m^Ig_I^{\frac{2}{3}}}$$
$$=\frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3}\left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{3}{2}\times\frac{5}{3}}\cdot\left(\frac{1}{6\pi^2}\right)^{\frac{5}{3}}\sum_I\frac{\int^{r_{max}}d^3r\left(\frac{f_I(r)}{r}\right)^{\frac{5}{2}}g_I^{\frac{5}{3}}}{m^Ig_I^{\frac{2}{3}}}$$
$$=\frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3m_1}\left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{5}{2}}\left(\frac{1}{6\pi^2}\right)^{\frac{5}{3}}\sum_I\int^{r_{max}}\alpha_Ig_I\left(\frac{f_I(r)}{r}\right)^{\frac{5}{2}}d^3r$$

where, $\alpha_I=\frac{m^1}{m^I}$

$$=\frac{3(6\pi^2\hbar^3)^{\frac{5}{3}}}{20\pi^2\hbar^3m_1}\left(\frac{2\times\frac{4}{3}\alpha_s}{a}\right)^{\frac{5}{2}}\left(\frac{1}{6\pi^2}\right)^{\frac{5}{3}}4\pi\sum_I\int^{r_{max}}drr^2\alpha_Ig_I\left(\frac{f_I(r)}{r}\right)^{\frac{5}{2}}$$

$$\longrightarrow T_1=\text{constant}\cdot\sum_I\int^{r_{max}}drr^2\alpha_Ig_I\left(\frac{f_I(r)}{r}\right)^{\frac{5}{2}} \tag{K3}$$

where,

$$\text{constant}=\frac{3\left(6\pi^2\hbar^3\right)^{\frac{5}{3}}}{20\pi^2\hbar^3m_1}\left(\frac{2\times\frac{4}{3}\alpha_s}{4}\right)^{\frac{5}{2}}\left(\frac{1}{6\pi^2}\right)^{\frac{5}{3}}\cdot4\pi$$

Put $r = Rx$

$$dr = Rdx$$

$$T_1 = \text{constant} \cdot \sum_I \int^{x_{max}} RdxR^2x^2\alpha_I \left(\frac{f_I(x)}{Rx}\right)^{\frac{5}{2}} g_I$$

$$= \text{constant} \cdot \sum_I \int^{x_{max}} \sqrt{R}\alpha_I g_I \frac{(f_I(x))^{\frac{5}{2}}}{\sqrt{x}} dx$$

$$= \text{constant} \cdot \sqrt{R} \sum_I \int^{x_{max}} \alpha_I g_I \frac{(f_I(x))^{\frac{5}{2}}}{\sqrt{x}} dx \tag{K4}$$

$$= C \cdot \sum_I \alpha_I g_I \int^{x_{max}} dx \frac{(f_I(x))^{\frac{5}{2}}}{\sqrt{x}} \tag{K5}$$

where, $C = \text{constant} \cdot \sqrt{R}$

$$\frac{T_1}{C} = \alpha_2 g_2 \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_0^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$\frac{T_1}{C} = \alpha_2 g_2 \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_0^{x_2} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_{x_2}^{\overline{x_2}} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$\frac{T_1}{C} = \alpha_2 g_2 \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 (\beta_{12})^{\frac{5}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} \sqrt{x} + \alpha_1 g_1 (\gamma_{1\overline{2}})^{\frac{5}{2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \alpha_1 g_1 \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}$$

Similarly,

$$\frac{T_2}{C} = \overline{\alpha_2 g_2} \int_0^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \overline{\alpha_2 g_2} \int_0^{x_2} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \overline{\alpha_2 g_2} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \overline{\alpha_2 g_2} (\beta_{\overline{2}\overline{2}})^{\frac{5}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \overline{\alpha_2 g_2} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

Adding,

$$\frac{T_1}{C} + \frac{T_2}{C} = \left[\alpha_1 g_1 (\beta_{12})^{\frac{5}{2}} + \alpha_2 g_2 + \overline{\alpha_2 g_2} (\beta_{\overline{2}\overline{2}})^{\frac{5}{2}}\right] \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \left[\alpha_1 g_1 (\gamma_{1\overline{2}})^{\frac{5}{2}} + \overline{\alpha_2 g_2}\right] \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$
$$+ \alpha_1 g_1 \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}$$

Substituting all terms and evaluating

$$\frac{T}{C} = \frac{4}{7}\left[\alpha_1 g_1 \left\{(\beta_{12})^{\frac{5}{2}} - (\gamma_{1\overline{2}})^{\frac{5}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}\right\} + \alpha_2 g_2\right] \sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7}\overline{\alpha_2 g_2}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7}\alpha_1 g_1 \sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}$$
$$+ \frac{5N_2}{21\eta}\left[\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 \frac{Q_2}{\overline{Q_2}} \left\{\alpha_1 g_1 (\gamma_{1\overline{2}})^{\frac{5}{2}} + \overline{\alpha_2 g_2}\right\} - \alpha_1 g_1 (\beta_{12})^{\frac{5}{2}} - \overline{\alpha_2 g_2}(\beta_{\overline{2}\overline{2}})^{\frac{5}{2}} - \alpha_2 g_2\right] \frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5}{21\eta}\left[\frac{(\gamma_{1\overline{2}})^2}{Q_1}\alpha_1 g_1 \left\{\overline{N_2 Q_2}\right.\right.$$
$$\left. + \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) N_2 Q_2 - (\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} N_2 \overline{Q_2}\right\} - \left\{\alpha_1 g_1 (\gamma_{1\overline{2}})^{\frac{5}{2}} + \overline{\alpha_2 g_2}\right\} \left\{\overline{N_2} + \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{N_2 Q_2}{\overline{Q_2}} - (\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} N_2\right\} \left.\right] \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\alpha_1 g_1}{21\eta}\left[N_1\right.$$
$$\left. + \gamma_{1\overline{2}} \frac{\overline{N_2 Q_2}}{Q_1} + (\gamma_{1\overline{2}})^{\frac{3}{2}} \left\{(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} N_2 - \overline{N_2}\right\} - (\beta_{12})^{\frac{3}{2}} N_2 - (\gamma_{1\overline{2}})(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} \frac{N_2 \overline{Q_2}}{Q_1} + \gamma_{1\overline{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{N_2 Q_2}{Q_1}\right] \frac{df_1(x)}{dx}\Big|_{x_1}$$

Calculation of K_{11}, K_{21}, K_{12}, $K_{\overline{2}1}$, K_{22}, $K_{\overline{2}2}$, $K_{\overline{22}}$

2.1 Calculation of $K_{\overline{22}}$

2.1.1 Appendix

$$\int_0^x dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

In region $0 < x < x_2$

$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_0^x dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$

$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_0^x dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2} \int_0^x dx x \frac{d^2 f_2(x)}{dx^2}$$

$$= \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2} \left[x\frac{df_2(x)}{dx} - f_2(x) \right]$$

In region $x_2 < x < \overline{x_2}$

$$= \int_0^x dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} + \int_{x_2}^x dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} + \int_{x_2}^x dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_0^{x_2} dx \frac{x}{Q_2} \frac{d^2 f_2(x)}{dx^2} + \int_{x_2}^x dx \frac{x}{\overline{Q_2}} \frac{d^2 \overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx x \frac{d^2 f_2(x)}{dx^2} + \frac{1}{\overline{Q_2}} \int_{x_2}^x dx x \frac{d^2 \overline{f_2(x)}}{dx^2}$$

$$= \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2} \left[x\frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} + \frac{1}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{x_2}^{x}$$

$$= \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2} \left[x\frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} + \frac{1}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{x} - \frac{1}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]_{x_2}$$

$$= \left\{ \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2} - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \right\} \left[x\frac{df_2(x)}{dx} - f_2(x) \right]_{x_2} + \frac{1}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]$$

$$= \frac{N_2 Q_2}{3\eta} \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2} - \frac{N_2 Q_2}{3\eta} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) + \frac{1}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]$$

$$= \frac{N_2}{3\eta} (\beta_{\overline{2}2})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta \overline{Q_2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) + \frac{1}{\overline{Q_2}} \left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right]$$

$$\int_x^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

In region $0 < x < x_2$

$$= \int_x^{x_2} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_x^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} + \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_x^{x_2} dx \frac{1}{Q_2} \frac{d^2 f_2(x)}{dx^2} + \int_{x_2}^{\overline{x_2}} dx \frac{1}{\overline{Q_2}} \frac{d^2 \overline{f_2(x)}}{dx^2} +$$

$$= \frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{Q_2}\frac{df_2(x)}{dx}\Big|_{x}^{x_2} + \frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{x_2}^{\overline{x_2}}$$

$$= \frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{Q_2}\frac{df_2(x)}{dx} + \frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{x_2}$$

In region $x_2 < x < \overline{x_2}$

$$= \int_{x}^{\overline{x_2}} dx \frac{1}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{x}^{\overline{x_2}}$$

$$= \frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{x}$$

$$\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$

$$= \int_{x_2}^{\overline{x_2}} dx \frac{x}{\overline{Q_2}}\frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{1}{\overline{Q_2}}\int_{x_2}^{\overline{x_2}} dx x\frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{1}{\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{x_2}^{\overline{x_2}}$$

$$= \frac{1}{\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}} - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}$$

$$= \frac{1}{\overline{Q_2}}\left[\frac{\overline{N_2}\overline{Q_2}}{3\eta} - (\beta_{\overline{22}})^{\frac{3}{2}}\frac{N_2\overline{Q_2}}{3\eta} + \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{N_2Q_2}{3\eta}\right] - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}\frac{N_2Q_2}{3\eta}$$

$$= \frac{\overline{N_2}}{3\eta} - (\beta_{\overline{22}})^{\frac{3}{2}}\frac{N_2}{3\eta} + \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{N_2Q_2}{3\eta\overline{Q_2}} - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}\frac{N_2Q_2}{3\eta}$$

$$= \frac{\overline{N_2}}{3\eta} - \frac{N_2}{3\eta}(\beta_{\overline{22}})^{\frac{3}{2}}$$

Now,

$$K_{\overline{22}} = \int_0^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^{x} dx'\sqrt{x'}(\overline{f_2(x')})^{\frac{3}{2}} + \int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\int_x^{\overline{x_2}} dx'\frac{(\overline{f_2(x')})^{\frac{3}{2}}}{\sqrt{x}}$$

$$K_{\overline{22}} = K_{\overline{22}}\cdot 1 + K_{\overline{22}}\cdot 2$$

$$K_{\overline{22}}\cdot 1 = \int_0^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^{x} dx'\sqrt{x'}(\overline{f_2(x')})^{\frac{3}{2}}$$

$$= \int_0^{x_2} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^{x} dx'\sqrt{x'}(\overline{f_2(x')})^{\frac{3}{2}} + \int_{x_2}^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^{x} dx'\sqrt{x'}(\overline{f_2(x')})^{\frac{3}{2}}$$

$$= \int_0^{x_2} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\left[\frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{Q_2}\left\{x\frac{df_2(x)}{dx} - f_2(x)\right\}\right] + \int_{x_2}^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\left[\frac{N_2}{3\eta}(\beta_{\overline{22}})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) + \frac{1}{\overline{Q_2}}\left\{x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right\}\right]$$

$$= \frac{(\beta_{\overline{22}})^3}{Q_2}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx} - \frac{(\beta_{\overline{22}})^3}{Q_2}\int_0^{x_2} dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta}(\beta_{\overline{22}})^{\frac{3}{2}}\int_{x_2}^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} - \frac{N_2Q_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\int_{x_2}^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$
$$+ \frac{1}{\overline{Q_2}}\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx} - \frac{1}{\overline{Q_2}}\int_{x_2}^{\overline{x_2}} dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$

$$K_{\overline{22}} \cdot 2 = \int_0^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \int_x^{\overline{x_2}} dx' \frac{(\overline{f_2(x')})^{\frac{3}{2}}}{\sqrt{x}}$$

$$= \int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \int_{x_2}^{\overline{x_2}} dx' \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \int_x^{\overline{x_2}} dx' \frac{(\overline{f_2(x')})^{\frac{3}{2}}}{\sqrt{x}}$$

$$= \int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \left[\frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx} + \frac{1}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \frac{df_2(x)}{dx}\bigg|_{x_2} + \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \right.$$
$$\left. \left[\frac{1}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{1}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{x_2} \right] \right]$$

$$= \frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{df_2(x)}{dx} + \frac{1}{\overline{Q_2}} \int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$
$$- \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{1}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{1}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}$$

$$= \frac{(\beta_{\overline{22}})^3}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{(\beta_{\overline{22}})^3}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx} + \frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$
$$- \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{1}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{1}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}$$

Adding $K_{\overline{22}} \cdot 1 + K_{\overline{22}} \cdot 2$

$$= -\frac{(\beta_{\overline{22}})^3}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta}(\beta_{\overline{22}})^{\frac{3}{2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} - \frac{N_2 Q_2}{3\eta\overline{Q_2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} - \frac{1}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$
$$+ \frac{(\beta_{\overline{22}})^3}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{(\beta_{\overline{22}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$
$$\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{1}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$

$$= -\frac{(\beta_{\overline{22}})^3}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{N_2 Q_2}{3\eta\overline{Q_2}^2} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$
$$+ \frac{N_2 Q_2}{3\eta\overline{Q_2}^2} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{1}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta Q_2}(\beta_{\overline{22}})^3 \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$
$$- \frac{N_2}{3\eta\overline{Q_2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{\overline{N_2}}{3\eta\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$

$$= -\frac{(\beta_{\overline{22}})^3}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{1}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{N_2}{3\eta\overline{Q_2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{N_2 Q_2}{3\eta\overline{Q_2}^2} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$
$$+ \frac{N_2 Q_2}{3\eta\overline{Q_2}^2} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{N_2}{3\eta Q_2}(\beta_{\overline{22}})^3 \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2}$$
$$+ \frac{\overline{N_2}}{3\eta\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$

$$= -\frac{(\beta_{\overline{22}})^3}{Q_2}\left[\frac{4}{7}\sqrt{x}\,(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5N_2}{21\eta}\,\frac{df_2(x)}{dx}\Big|_{x_2}\right] - \frac{1}{\overline{Q_2}}\left[\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7}\sqrt{x}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5}{21\eta}\Big\{\overline{N_2}$$

$$- N_2(\beta_{\overline{22}})^{\frac{3}{2}} + \frac{N_2Q_2}{\overline{Q_2}}\beta_{\overline{22}}\Big\}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5}{21\eta}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{N_2Q_2}{\overline{Q_2}}\,\frac{df_2(x)}{dx}\Big|_{x_2}\Big] - \frac{N_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{N_2Q_2}{3\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2Q_2}{3\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(\beta_{\overline{22}})^3\,\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$- \frac{N_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta\overline{Q_2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$= -\frac{4}{7Q_2}(\beta_{\overline{22}})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{5N_2}{21\eta Q_2}(\beta_{\overline{22}})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{4}{7\overline{Q_2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}$$

$$+ \frac{5\overline{N_2}}{21\eta\overline{Q_2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{N_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2Q_2}{3\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2Q_2}{3\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(\beta_{\overline{22}})^3\,\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$+ \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta\overline{Q_2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$= \frac{4}{7\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}(\beta_{\overline{22}})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7\overline{Q_2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{5N_2}{21\eta Q_2}(\beta_{\overline{22}})^3\,\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2Q_2}{3\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta Q_2}(\beta_{\overline{22}})^3\,\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$- \frac{N_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$- \frac{N_2Q_2}{3\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{22}})^{\frac{3}{2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta\overline{Q_2}}\,\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

2.2 Calculation of K_{22}

$$K_{22} = \int_0^{x_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}dx\int_0^{x}\sqrt{x'}(f_2(x'))^{\frac{3}{2}}dx' + \int_0^{x_2}\sqrt{x}(f_2(x))^{\frac{3}{2}}dx\int_x^{x_2}\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}dx'$$

$$= \int_0^{x_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}dx\left[\frac{1}{Q_2}\left(x'\frac{df_2(x')}{dx'} - f_2(x')\right)\Big|_x\right] + \int_0^{x_2}\sqrt{x}(f_2(x))^{\frac{3}{2}}dx\frac{1}{Q_2}\int_x^{x_2}\frac{d^2f_2(x')}{dx'^2}dx'$$

$$= \int_0^{x_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}dx\cdot\frac{1}{Q_2}\left(x'\frac{df_2(x')}{dx'} - f_2(x')\right)\Big|_x + \frac{1}{Q_2}\int_0^{x_2}\sqrt{x}(f_2(x))^{\frac{3}{2}}dx\left(\frac{df_2(x')}{dx'}\Big|_{x_2} - \frac{df_2(x')}{dx'}\Big|_x\right)$$

$$= \int_0^{x_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}dx\cdot\frac{1}{Q_2}\left(x'\frac{df_2(x')}{dx'} - f_2(x')\right)\Big|_x + \frac{1}{Q_2}\frac{df_2(x')}{dx'}\Big|_{x_2}\cdot\frac{N_2}{3\eta} - \frac{1}{Q_2}\int_0^{x_2}\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x')}{dx'}\Big|_x dx$$

$$= \frac{1}{Q_2}\int_0^{x_2}dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx} - \frac{1}{Q_2}\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$= -\frac{1}{Q_2}\int_0^{x_2}\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}dx + \frac{N_2}{3\eta}\frac{1}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$= -\frac{1}{Q_2}\left\{\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5N_2}{21\eta}\frac{df_2(x)}{x}\Big|_{x_2}\right\} + \frac{1}{Q_2}\frac{N_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$\boxed{K_2 = -\frac{4}{7Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4N_2}{7\eta Q_2}\frac{df_2(x)}{dx}\Big|_{x_2}}$$

2.3 Calculation of K_{12}

$$K_{12} = \int_0^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}})}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_0^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$
$$= K_{12} \cdot 1 + K_{12} \cdot 2$$

$$K_{12} \cdot 1 = \int_0^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}$$
$$= \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} + \int_{x_2}^{\overline{x_2}} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}$$
$$+ \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}$$

2.3.1 Taking 1st term

$$\int_0^{x_2} dx \frac{(f(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}$$
$$= (\beta_{12})^{\frac{3}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}$$
$$\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \Big(x' \frac{df_2(x')}{dx'} - f_2(x') \Big) \Big|_x$$
$$= \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx (f_2(x))^{\frac{3}{2}} \sqrt{x} \frac{df_2(x')}{dx'} \Big|_x - \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

2.3.2 Taking 2nd term

$$= \int_{x_2}^{\overline{x_2}} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}} \qquad for \qquad [x_2 < x < \overline{x_2}]$$
$$= (\gamma_{1\overline{2}})^{\frac{3}{2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \Big\{ \underbrace{\int_0^{x_2} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}}_{\text{Using NC}} + \underbrace{\int_{x_2}^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}}_{[f_2(x)=0]} \Big\}$$
$$= (\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{N_2}{3\eta} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

2.3.3 Taking 3rd term

$$\int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}$$
$$= \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \Big\{ \underbrace{\int_0^{x_2} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}}_{\text{Using NC}} + \underbrace{\int_{x_2}^{\overline{x_2}} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}}_{[f_2(x)=0]} + \underbrace{\int_{\overline{x_2}}^{x} dx' \sqrt{x'} (f_2(x'))^{\frac{3}{2}}}_{[f_2(x)=0]} \Big\}$$
$$= \frac{N_2}{3\eta} \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$= \frac{N_2}{3\eta} \frac{1}{Q_1} \Big[\frac{df_1(x)}{dx} \Big|_{x_1} - \frac{df_1(x)}{dx} \Big|_{\overline{x_2}} \Big]$$

$$K_{12} \cdot 2 = \int_0^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= \int_0^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \underbrace{\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}}_{[x_2<x<\overline{x_2}]} \underbrace{\int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}}_{[f_2(x)=0]}$$

$$+ \underbrace{\int_{\overline{x_2}}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}}_{[\overline{x_2}<x<x_1]} \underbrace{\int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}}_{[f_2(x)=0]}$$

$$= \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x')}{dx'}\bigg|_{x_2} - \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x')}{dx'}\bigg|_x$$

On adding 1st, 2nd, 3rd term and $K_{12} \cdot 2$

$$= \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx(f_2(x))^{\frac{3}{2}} \sqrt{x} \frac{df_2(x')}{dx'}\bigg|_x - \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + (\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{N_2}{3\eta} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$
$$+ \frac{N_2}{3\eta} \frac{1}{Q_1} \left[\frac{df_1(x')}{dx'}\bigg|_{x_1} - \frac{df_1(x')}{dx'}\bigg|_{\overline{x_2}} \right] + \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x')}{dx'}\bigg|_{x_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$
$$- \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_x$$

$$= -\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + (\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{N_2}{3\eta} \int_{x_2}^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta} \frac{1}{Q_1} \left[\frac{df_1(x'}{dx'}\bigg|_{x_1} - \frac{df_1(x')}{dx'}\bigg|_{\overline{x_2}} \right]$$
$$+ \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x')}{dx'}\bigg|_{x_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}$$

$$= -\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \left[\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\bigg|_{x_2} - \frac{5}{7Q_2} \frac{df_2(x)}{dx}\bigg|_{x_2} \cdot \frac{N_2 Q_2}{3\eta} \right] + (\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{N_2}{3\eta} \left[\frac{1}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \frac{df_2(x)}{dx}\bigg|_{x_2} \right]$$
$$+ \frac{N_2}{3\eta Q_1} \left[\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{df_1(x)}{dx}\bigg|_{\overline{x_2}} \right] + \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \cdot \frac{N_2}{3\eta} \frac{df_2(x)}{dx}\bigg|_{x_2}$$

$$\boxed{\begin{aligned} K_{12} = &- \frac{4}{7Q_2}(\beta_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\bigg|_{x_2} + \frac{N_2}{3\eta}\left\{ \frac{5}{7Q_2}(\beta_{12})^{\frac{3}{2}} + \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \right\} \frac{df_2(x)}{dx}\bigg|_{x_2} \\ &+ \frac{N_2}{3\eta}\left\{ \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\gamma_{1\overline{2}}}{Q_1} \right\} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{N_2}{3\eta Q_1} \frac{df_1(x)}{dx}\bigg|_{x_1} \end{aligned}}$$

2.4 Calculation of K_{21}

$$K_{21} = \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx'\sqrt{x'}(f_1(x'))^{\frac{3}{2}} + \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$
$$= K_{21} \cdot 1 + K_{21} \cdot 2$$

$$K_{21} \cdot 1 = \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx'\sqrt{x'}(f_1(x'))^{\frac{3}{2}}$$
$$= (\beta_{12})^{\frac{3}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx'\sqrt{x'}(f_2(x'))^{\frac{3}{2}}$$
$$= \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[x' \frac{df_2(x')}{dx'} - f_2(x') \right]\bigg|_x$$
$$= \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x')}{dx'}\bigg|_x - \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$K_{21} \cdot 2 = \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \left\{ \int_x^{x_2} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{\overline{x_2}} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \right\}$$

$$= \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \left\{ \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x')}{dx'}\Big|_{x_2} - \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x')}{dx'}\Big|_{x} + \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x')}}{dx'}\Big|_{x_2} \right.$$
$$\left. + \frac{1}{Q_1} \frac{df_1(x')}{dx'}\Big|_{x_1} - \frac{1}{Q_1} \frac{df_1(x')}{dx'}\Big|_{\overline{x_2}} \right\}$$

$$= \left\{ \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x')}{dx'}\Big|_{x_2} + \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x')}}{dx'}\Big|_{\overline{x_2}} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x')}}{dx'}\Big|_{x_2} + \frac{1}{Q_1} \frac{df_1(x')}{dx'}\Big|_{x_1} - \frac{1}{Q_1} \frac{df_1(x')}{dx'}\Big|_{\overline{x_2}} \right\}$$
$$\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} - \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} \frac{df_2(x')}{dx'}\Big|_x \sqrt{x}(f_2(x))^{\frac{3}{2}} dx$$

Finally on adding

$$K_{21} = K_{21} \cdot 1 + K_{21} \cdot 2$$

$$= -\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \left\{ \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x')}{dx'}\Big|_{x_2} + \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x')}}{dx'}\Big|_{\overline{x_2}} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x')}}{dx'}\Big|_{x_2} \right.$$
$$\left. + \frac{1}{Q_1} \frac{df_1(x')}{dx'}\Big|_{x_1} - \frac{1}{Q_1} \frac{df_1(x')}{dx'}\Big|_{\overline{x_2}} \right\} \frac{N_2}{3\eta}$$

$$= -\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \left[\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5N_2}{21\eta Q_2} \frac{df_2(x)}{dx}\Big|_{x_2} \right] + \frac{N_2}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left[\frac{1}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \frac{df_2(x)}{dx}\Big|_{x_2} \right]$$
$$+ \frac{N_2}{3\eta Q_1} \left[\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{df_1(x)}{dx}\Big|_{\overline{x_2}} \right] + \frac{N_2}{3\eta Q_2}(\beta_{12})^{\frac{3}{2}} \frac{df_2(x)}{dx}\Big|_{x_2}$$

$$\boxed{\begin{aligned} K_{21} = &-\frac{4}{7Q_2}(\beta_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{N_2}{3\eta}\left\{ \frac{5}{7Q_2}(\beta_{12})^{\frac{3}{2}} + \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \right\} \frac{df_2(x)}{dx}\Big|_{x_2} \\ &+ \frac{N_2}{3\eta}\left\{ \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{\gamma_{1\overline{2}}}{Q_1} \right\} \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2}{3\eta Q_1} \frac{df_1(x)}{dx}\Big|_{x_1} \end{aligned}}$$

Hence,

$$\boxed{K_{12} = K_{21} \qquad \text{[proved]}}$$

2.5 $K_{\overline{2}2}$

$$K_{\overline{2}2} = \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} dx \int_0^{x} \sqrt{x'}(f_2(x'))^{\frac{3}{2}} dx' + \int_0^{\overline{x_2}} \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} dx \int_x^{x_2} \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} dx'$$
$$= K_{\overline{2}2} \cdot 1 + K_{\overline{2}2} \cdot 2$$

$$K_{\overline{2}2} \cdot 1 = \int_0^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} dx \int_0^{x} \sqrt{x'}(f_2(x'))^{\frac{3}{2}} dx'$$

$$= \int_0^{x_2} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} dx \int_0^{x} \sqrt{x'}(f_2(x'))^{\frac{3}{2}} dx' + \int_{x_2}^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} dx \int_0^{x} \sqrt{x'}(f_2(x'))^{\frac{3}{2}} dx'$$

$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_0^{x_2} \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} dx \int_0^{x} \sqrt{x'}(f_2(x'))^{\frac{3}{2}} dx + \int_{x_2}^{\overline{x_2}} \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} dx \left\{ \underbrace{\int_0^{x_2} \sqrt{x'}(f_2(x'))^{\frac{3}{2}} dx'}_{\text{Using NC}} \right.$$
$$\left. + \underbrace{\int_{x_2}^{x} \sqrt{x'}(f_2(x'))^{\frac{3}{2}} dx'}_{[f_2(x)=0]} \right\}$$

$$= \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}dx\left(x\frac{df_2(x)}{dx}-f_2(x)\right)\Big|_x+\frac{N_2}{3\eta}\left[\frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{x_2}\right]$$

$$K_{\overline{2}2}\cdot 2=\int_0^{\overline{x_2}}\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}dx\int_x^{x_2}\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}dx'$$

$$=\int_0^{x_2}\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}dx\int_x^{x_2}\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}dx'+\int_{x_2}^{\overline{x_2}}\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}dx\underbrace{\left(-\int_{x_2}^{x}\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}}dx'\right)}_{[f_2(x)=0]}$$

$$=\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\left(\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{df_2(x)}{dx}\Big|_x\right)$$

$$=\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}$$

Finally,

$$K_{\overline{2}2}=K_{\overline{2}2}\cdot 1+K_{\overline{2}2}\cdot 2$$

$$=\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}dx\left(x\frac{df_2(x)}{dx}-f_2(x)\right)\Big|_x+\frac{N_2}{3\eta}\left[\frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{x_2}\right]$$
$$+\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}\sqrt{x}(f_2(x))^{\frac{3}{2}}dx\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}$$

$$=-\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}dx+\frac{N_2}{3\eta}\frac{1}{\overline{Q_2}}\left[\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{d\overline{f_2(x)}}{dx}\Big|_{x_2}\right]+\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\frac{N_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2}$$

$$=-\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}\left\{\frac{4}{7}\sqrt{x_2}(f_2(x))^{\frac{5}{2}}-\frac{5N_2}{21\eta}\frac{df_2(x)}{dx}\Big|_{x_2}\right\}+\left\{\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}-\frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}\right\}\frac{N_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{N_2}{3\eta}\frac{1}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$

$$\boxed{K_{\overline{2}2}=-\frac{4}{7Q_2}(\beta_{\overline{2}2})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}+\frac{N_2}{3\eta}\left\{\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_2}-\frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}+\frac{5}{7Q_2}(\beta_{\overline{2}2})^{\frac{3}{2}}\right\}\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{N_2}{3\eta\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}}$$

Calculation of $K_{\overline{2}1}$

$$K_{\overline{2}1}=\int_0^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^{x}dx'\sqrt{x'}(f_1(x'))^{\frac{3}{2}}+\int_0^{\overline{x_2}}dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\int_x^{x_1}dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{128}$$

$$K_{\overline{2}1}=K_{\overline{2}1}\cdot 1+K_{\overline{2}1}\cdot 2 \tag{129}$$

At first,

$$K_{\overline{2}1}\cdot 1=\int_0^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^{x}dx'\sqrt{x'}(f_1(x'))^{\frac{3}{2}} \tag{130}$$

$$=\int_0^{x_2}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^{x}dx'\sqrt{x'}(f_1(x'))^{\frac{3}{2}}+\int_{x_2}^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\int_0^{x}dx'\sqrt{x'}(f_1(x'))^{\frac{3}{2}}$$

$$=(\beta_{\overline{2}2})^{\frac{3}{2}}\int_0^{x_2}dx\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\left\{\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2}\left(x\frac{df_2(x)}{dx}-f_2(x)\right)\right\}+\int_{x_2}^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\left[\int_0^{x_2}dx'\sqrt{x'}(f_1(x'))^{\frac{3}{2}}+\int_{x_2}^{x}dx'\sqrt{x'}(f_1(x'))^{\frac{3}{2}}\right]$$

$$=\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}}{Q_2}\left[\int_0^{x_2}dx\frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}}\cdot x\frac{df_2(x)}{dx}-\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}\right]+\int_{x_2}^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}\left[\frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}}+\frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left(x\frac{d\overline{f_2(x)}}{dx}-\overline{f_2(x)}\right)\right.$$
$$\left.-\frac{N_2Q_2}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right]$$

$$=\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}dx(f_2(x))^{\frac{3}{2}}\sqrt{x}\frac{df_2(x)}{dx}-\frac{(\beta_{\overline{2}2})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}}{Q_2}\int_0^{x_2}dx\frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}}+\frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}}\int_{x_2}^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$
$$+\frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_{x_2}^{\overline{x_2}}dx(\overline{f_2(x)})^{\frac{3}{2}}\sqrt{x}\frac{d\overline{f_2(x)}}{dx}-\frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\int_{x_2}^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}-\frac{N_2Q_2}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\int_{x_2}^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

The first term and fourth term gets cancelled with $K_{\overline{21}} \cdot 2$

$$= \frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} - \frac{(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{N_2 Q_2}{3\eta \overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$

$$= \frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}} \int_{x_2}^{\overline{x_2}} dx \frac{1}{\overline{Q_2}} \frac{d^2\overline{f_2(x)}}{dx^2} - \frac{(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}}{Q_2} \left[\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}} \Big|_{x_2} - \frac{5N_2}{21\eta} \frac{df_2(x)}{dx} \Big|_{x_2} \right] - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} \right.$$
$$- \frac{4}{7}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}} \sqrt{x}(f_2(x))^{\frac{5}{2}} \Big|_{x_2} - \frac{5}{7\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \left\{ \frac{\overline{N_2 Q_2}}{3\eta} + \frac{N_2 Q_2}{3\eta}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{N_2 \overline{Q_2}}{3\eta}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} \right\}$$
$$\left. + \frac{5}{7\overline{Q_2}} \frac{N_2 Q_2}{3\eta}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 \frac{df_2(x)}{dx} \Big|_{x_2} \right] - \frac{N_2 Q_2}{3\eta \overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \int_{x_2}^{\overline{x_2}} dx \frac{1}{\overline{Q_2}} \frac{d^2\overline{f_2(x)}}{dx^2}$$

$$= \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{4}{7Q_2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}} \sqrt{x}(f_2(x))^{\frac{5}{2}} \Big|_{x_2} + \frac{5N_2}{21\eta Q_2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}$$
$$\frac{df_2(x)}{dx} \Big|_{x_2} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left[\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} - \frac{4}{7}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}} \sqrt{x}(f_2(x))^{\frac{5}{2}} \Big|_{x_2} - \frac{5\overline{N_2}}{21\eta} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{5N_2 Q_2}{21\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} \right.$$
$$\left. + \frac{5N_2}{21\eta}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{5N_2 Q_2}{21\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 \frac{df_2(x)}{dx} \Big|_{x_2} \right] - \frac{N_2 Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}}$$
$$+ \frac{N_2 Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$= \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{4}{7Q_2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}} \sqrt{x}(f_2(x))^{\frac{5}{2}} \Big|_{x_2}$$
$$+ \frac{5N_2}{21\eta Q_2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}} + \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}} \sqrt{x}(f_2(x))^{\frac{5}{2}} \Big|_{x_2}$$
$$+ \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{5N_2 Q_2}{21\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}}$$
$$- \frac{5N_2 Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 (\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{N_2 Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{N_2 Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 \frac{df_2(x)}{dx} \Big|_{x_2}$$

$$= \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}} \sqrt{x}(f_2(x))^{\frac{5}{2}} \Big|_{x_2} - \frac{4}{7Q_2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}} \sqrt{x}(f_2(x))^{\frac{5}{2}} \Big|_{x_2} - \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}} \Big|_{\overline{x_2}}$$
$$+ \frac{5N_2}{21\eta Q_2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}} \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{df_2(x)}{dx} \Big|_{x_2} - \frac{5N_2 Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 (\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{df_2(x)}{dx} \Big|_{x_2}$$
$$+ \frac{N_2 Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 \frac{df_2(x)}{dx} \Big|_{x_2} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}}$$
$$+ \frac{5N_2 Q_2}{21\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}} - \frac{N_2 Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) \frac{d\overline{f_2(x)}}{dx} \Big|_{\overline{x_2}}$$

At second,

$$K_{\overline{21}} \cdot 2 = \int_0^{\overline{x_2}} dx \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{131}$$

$$= \int_0^{x_2} dx \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{\overline{x_2}} dx \sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \int_x^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

$$= \int_0^{x_2} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \left\{ \int_x^{x_2} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{\overline{x_2}} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \right\} + \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}$$
$$\left\{ \int_x^{\overline{x_2}} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \right\}$$
$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \left\{ (\beta_{12})^{\frac{3}{2}} \int_x^{x_2} dx' \frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} + (\gamma_{1\overline{2}})^{\frac{3}{2}} \int_{x_2}^{\overline{x_2}} dx' \frac{(\overline{f_2(x')})^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \right\}$$
$$+ \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \left\{ (\gamma_{1\overline{2}})^{\frac{3}{2}} \int_x^{\overline{x_2}} dx' \frac{(\overline{f_2(x')})^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_1} dx' \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \right\}$$
$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \left[\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_x^{x_2} dx' \frac{d^2 f_2(x')}{dx'^2} + \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx' \frac{d^2\overline{f_2(x')}}{dx'^2} + \frac{1}{Q_1} \int_{\overline{x_2}}^{x_1} dx' \frac{d^2 f_1(x')}{dx'^2} \right]$$
$$+ \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(f_2(x'))^{\frac{3}{2}} \left[\frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_x^{\overline{x_2}} dx' \frac{d^2\overline{f_2(x')}}{dx'^2} + \frac{1}{Q_1} \int_{\overline{x_2}}^{x_1} dx' \frac{d^2 f_1(x')}{dx'^2} \right]$$

$$= (\beta_{\overline{2}2})^{\frac{3}{2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \left[\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \frac{df_2(x)}{dx} + \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \frac{df_2(x)}{dx}\bigg|_{x_2} \right.$$
$$\left. + \frac{1}{Q_1} \frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\gamma_{1\overline{2}}}{Q_1} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \right] + \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \left[\frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_1} \frac{df_1(x)}{dx}\bigg|_{x_1} \right.$$
$$\left. - \frac{\gamma_{1\overline{2}}}{Q_1} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \right]$$
$$= \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}}{Q_2} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx}$$
$$+ \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{2}2})^{\frac{3}{2}}}{\overline{Q_2}} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2}$$
$$+ \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_1} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}}{Q_1} \int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$
$$- \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_1} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\gamma_{1\overline{2}}}{Q_1} \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$

The second term and eighth term gets cancelled with $K_{\overline{2}1} \cdot 1$

$$= \frac{N_2}{3\eta Q_2}(\beta_{\overline{2}2})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\beta_{\overline{2}2})^{\frac{3}{2}} \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{N_2}{3\eta Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}}$$
$$\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{N_2}{3\eta Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_1} \frac{df_1(x)}{dx}\bigg|_{x_1}$$
$$- \frac{N_2}{3\eta Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}} \frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\overline{N_2}}{3\eta Q_1}\gamma_{1\overline{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{N_2}{3\eta Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}}$$

$$= \frac{N_2}{3\eta}(\beta_{\overline{2}2})^{\frac{3}{2}} \left\{ \frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} \right\} \frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{\overline{N_2}}{3\eta}\gamma_{1\overline{2}} \left\{ \frac{(\gamma_{1\overline{2}})^{\frac{1}{2}}}{\overline{Q_2}} - \frac{1}{Q_1} \right\} \frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_1} \frac{df_1(x)}{dx}\bigg|_{x_1} \quad (132)$$

Therefore, $K_{\overline{2}1} = K_{\overline{2}1} \cdot 1 + K_{\overline{2}1} \cdot 2$

$$
\begin{aligned}
&= \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}(\beta_{\overline{2}2})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} \\
&+ \frac{5N_2}{21\eta Q_2}(\beta_{\overline{2}2})^{\frac{3}{2}}(\beta_{12})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} \\
&+ \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&+ \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&+ \frac{N_2}{3\eta}(\beta_{\overline{2}2})^{\frac{3}{2}}\left\{\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{\overline{N_2}}{3\eta}\gamma_{1\overline{2}}\left\{\frac{(\gamma_{1\overline{2}})^{\frac{1}{2}}}{\overline{Q_2}} - \frac{1}{Q_1}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
$$

Calculation of K_{11}

$$K_{11} = \int_0^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_0^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \int_x^{x_1} dx \frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{133}$$

$$K_{11} = K_{11} \cdot 1 + K_{11} \cdot 2 \tag{134}$$

At first,

$$K_{11} \cdot 1 = \int_0^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} \tag{135}$$

$$= \int_0^{x_2} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_{x_2}^{\overline{x_2}} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}} + \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \int_0^{x} dx' \sqrt{x'} (f_1(x'))^{\frac{3}{2}}$$

$$= (\beta_{12})^{\frac{3}{2}} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2} \left\{ x \frac{df_2(x)}{dx} - f_2(x) \right\} \right] + (\gamma_{1\overline{2}})^{\frac{3}{2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{N_2}{3\eta} (\beta_{12})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta \overline{Q_2}} (\gamma_{1\overline{2}})^{\frac{3}{2}} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}} \right) \right.$$
$$+ \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} \left\{ x \frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)} \right\} \Bigg] + \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \left[\frac{N_2}{3\eta} (\beta_{12})^{\frac{3}{2}} + \frac{\overline{N_2}}{3\eta} (\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_1} \gamma_{1\overline{2}} - \frac{N_2}{3\eta} (\gamma_{1\overline{2}})^{\frac{3}{2}} (\beta_{\overline{2}2})^{\frac{3}{2}} \right.$$
$$\left. - \frac{N_2 Q_2}{3\eta Q_1} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}} \right) \gamma_{1\overline{2}} + \frac{N_2 \overline{Q_2}}{3\eta Q_1} (\beta_{\overline{2}2})^{\frac{3}{2}} \gamma_{1\overline{2}} + \frac{1}{Q_1} \left\{ x \frac{df_1(x)}{dx} - f_1(x) \right\} \right]$$

$$= \frac{(\beta_{12})^3}{Q_2} \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx} - \frac{(\beta_{12})^3}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} + \frac{N_2}{3\eta} (\beta_{12})^{\frac{3}{2}} (\gamma_{1\overline{2}})^{\frac{3}{2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}}$$
$$- \frac{N_2 Q_2}{3\eta \overline{Q_2}} (\gamma_{1\overline{2}})^3 \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}} \right) \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} - \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}}$$
$$+ \left\{ \frac{N_2}{3\eta} (\beta_{12})^{\frac{3}{2}} + \frac{\overline{N_2}}{3\eta} (\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_1} \gamma_{1\overline{2}} - \frac{N_2}{3\eta} (\gamma_{1\overline{2}})^{\frac{3}{2}} (\beta_{\overline{2}2})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta Q_1} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}} \right) \gamma_{1\overline{2}} + \frac{N_2 \overline{Q_2}}{3\eta Q_1} \gamma_{1\overline{2}} (\beta_{\overline{2}2})^{\frac{3}{2}} \right\} \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$
$$+ \frac{1}{Q_1} \int_{\overline{x_2}}^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \frac{df_1(x)}{dx} - \frac{1}{Q_1} \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}}$$

$$= \frac{(\beta_{12})^3}{Q_2} \int_0^{x_2} dx \sqrt{x} (f_2(x))^{\frac{3}{2}} \frac{df_2(x)}{dx} + \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \sqrt{x} (\overline{f_2(x)})^{\frac{3}{2}} \frac{d\overline{f_2(x)}}{dx} + \frac{1}{Q_1} \int_{\overline{x_2}}^{x_1} dx \sqrt{x} (f_1(x))^{\frac{3}{2}} \frac{df_1(x)}{dx}$$
$$- \frac{(\beta_{12})^3}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{1}{Q_1} \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \left\{ \frac{N_2}{3\eta} (\beta_{12})^{\frac{3}{2}} (\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta \overline{Q_2}} (\gamma_{1\overline{2}})^3 \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}} \right) \right\}$$
$$\int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + \left\{ \frac{N_2}{3\eta} (\beta_{12})^{\frac{3}{2}} + \frac{\overline{N_2}}{3\eta} (\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_1} \gamma_{1\overline{2}} - \frac{N_2}{3\eta} (\gamma_{1\overline{2}})^{\frac{3}{2}} (\beta_{\overline{2}2})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta Q_1} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}} \right) \gamma_{1\overline{2}} + \frac{N_2 \overline{Q_2}}{3\eta Q_1} (\beta_{\overline{2}2})^{\frac{3}{2}} \gamma_{1\overline{2}} \right\}$$
$$\int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}}$$

The first, second and third terms get cancelled with $K_{11} \cdot 2$,

$$= - \frac{(\beta_{12})^3}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{1}{Q_1} \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} + \left\{ \frac{N_2}{3\eta} (\beta_{12})^{\frac{3}{2}} (\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta \overline{Q_2}} (\gamma_{1\overline{2}})^3 \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}} \right) \right\}$$
$$\int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} + \left\{ \frac{N_2}{3\eta} (\beta_{12})^{\frac{3}{2}} + \frac{\overline{N_2}}{3\eta} (\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2 Q_2}}{3\eta Q_1} \gamma_{1\overline{2}} - \frac{N_2}{3\eta} (\gamma_{1\overline{2}})^{\frac{3}{2}} (\beta_{\overline{2}2})^{\frac{3}{2}} - \frac{N_2 Q_2}{3\eta Q_1} \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}} \right) \gamma_{1\overline{2}} + \frac{N_2 \overline{Q_2}}{3\eta Q_1} (\beta_{\overline{2}2})^{\frac{3}{2}} \gamma_{1\overline{2}} \right\}$$
$$\int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{136}$$

For convenience, above equation is evaluated in three sections separately.

- Section 1

$$- \frac{(\beta_{12})^3}{Q_2} \int_0^{x_2} dx \frac{(f_2(x))^{\frac{5}{2}}}{\sqrt{x}} - \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}} \int_{x_2}^{\overline{x_2}} dx \frac{(\overline{f_2(x)})^{\frac{5}{2}}}{\sqrt{x}} - \frac{1}{Q_1} \int_{\overline{x_2}}^{x_1} dx \frac{(f_1(x))^{\frac{5}{2}}}{\sqrt{x}} \tag{137}$$

Substituting values of above integration from appendix

$$
\begin{aligned}
&= -\frac{(\beta_{12})^3}{Q_2}\left[\frac{4}{7}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5N_2}{21\eta}\frac{df_2(x)}{dx}\Big|_{x_2}\right] - \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}}\left[\frac{4}{7}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{5}{7\overline{Q_2}}\left\{\frac{\overline{N_2Q_2}}{3\eta}\right.\right.\\
&\quad + \frac{N_2Q_2}{3\eta}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \left.\left.\frac{N_2\overline{Q_2}}{3\eta}(\beta_{\overline{2}2})^{\frac{3}{2}}\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5}{7\overline{Q_2}}\frac{N_2Q_2}{3\eta}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2}\right] - \frac{1}{Q_1}\left[\frac{4}{7}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}\right.\\
&\quad - \frac{4}{7}(\gamma_{1\overline{2}})^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{5}{7Q_1}\left[\frac{N_1Q_1}{3\eta} + \frac{\overline{N_2Q_2}}{3\eta}\gamma_{1\overline{2}} - \frac{\overline{N_2}Q_1}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}} + \frac{N_2Q_1}{3\eta}\{(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}} - (\beta_{12})^{\frac{3}{2}}\}\right.\\
&\quad \left.\left.\left. - \frac{N_2\overline{Q_2}}{3\eta}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}} + \frac{N_2Q_2}{3\eta}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right]\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5}{7Q_1}(\gamma_{1\overline{2}})^2\left[\frac{\overline{N_2Q_2}}{3\eta} + \frac{N_2Q_2}{3\eta}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{N_2\overline{Q_2}}{3\eta}(\beta_{\overline{2}2})^{\frac{3}{2}}\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right]
\end{aligned}
$$

$$
\begin{aligned}
&= -\frac{4}{7}\frac{(\beta_{12})^3}{Q_2}\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{5N_2}{21\eta}\frac{(\beta_{12})^3}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{4}{7}\frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{4}{7\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(\gamma_{1\overline{2}})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}\\
&\quad + \frac{5}{7\overline{Q_2}}\frac{\overline{N_2Q_2}}{3\eta}\frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5}{7\overline{Q_2}}\frac{N_2Q_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5}{7\overline{Q_2}}\frac{N_2\overline{Q_2}}{3\eta\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\\
&\quad - \frac{5}{7\overline{Q_2}}\frac{N_2Q_2}{3\eta\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2(\gamma_{1\overline{2}})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \frac{4}{7Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} + \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_1Q_1}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1}\\
&\quad + \frac{5}{7Q_1}\frac{\overline{N_2Q_2}}{3\eta Q_1}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{1}{Q_1}\frac{5}{7Q_1}\frac{\overline{N_2}Q_1}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2Q_1}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}\\
&\quad - \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2Q_1}{3\eta}(\beta_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{1}{Q_1}\frac{5}{7Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{N_2\overline{Q_2}}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2Q_2}{3\eta}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\Big|_{x_1}\\
&\quad - \frac{1}{Q_1}\frac{5}{7Q_1}\frac{\overline{N_2Q_2}}{3\eta}(\gamma_{1\overline{2}})^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2Q_2}{3\eta}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_1}\frac{5}{7Q_1}\frac{N_2\overline{Q_2}}{3\eta}(\gamma_{1\overline{2}})^2(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{4}{7\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(\gamma_{1\overline{2}})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}(\beta_{12})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}\\
&\quad - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \frac{5N_2}{21\eta Q_2}(\beta_{1\overline{2}})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2(\gamma_{1\overline{2}})^3\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\\
&\quad + \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}(\gamma_{1\overline{2}})^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5N_2Q_2}{21\eta{Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\\
&\quad \frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5N_2\overline{Q_2}}{21\eta{Q_1}^2}(\gamma_{1\overline{2}})^2(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5\overline{N_2}}{21\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}\\
&\quad + \frac{5N_2}{21\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5N_2}{21\eta Q_1}(\beta_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5N_2\overline{Q_2}}{21\eta{Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5N_2Q_2}{21\eta{Q_1}^2}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
\tag{138}
$$

- Section 2

$$
\left\{\frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right\}\int_{x_2}^{\overline{x_2}}dx\frac{(\overline{f_2(x)})^{\frac{3}{2}}}{\sqrt{x}} \tag{139}
$$

$$
= \frac{1}{\overline{Q_2}}\left\{\frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right\}\int_{x_2}^{\overline{x_2}}dx\frac{d^2\overline{f_2(x)}}{dx^2}
$$

$$
= \frac{1}{\overline{Q_2}}\left\{\frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right\}\left[\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{d\overline{f_2(x)}}{dx}\Big|_{x_2}\right]
$$

$$= \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2}$$
$$+ \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\bigg|_{x_2} \tag{140}$$

- Section 3

$$\left\{\frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}}+\frac{\overline{N_2}}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta Q_1}\gamma_{1\overline{2}}-\frac{N_2}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta Q_1}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}+\frac{N_2\overline{Q_2}}{3\eta Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\right\}\int_{\overline{x_2}}^{x_1}dx\frac{(f_1(x))^{\frac{3}{2}}}{\sqrt{x}} \tag{141}$$

$$=\frac{1}{Q_1}\left\{\frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}}+\frac{\overline{N_2}}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta Q_1}\gamma_{1\overline{2}}-\frac{N_2}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta Q_1}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}+\frac{N_2\overline{Q_2}}{3\eta Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\right\}\int_{\overline{x_2}}^{x_1}dx\frac{d^2f_1(x)}{dx^2}$$

$$=\frac{1}{Q_1}\left\{\frac{N_2}{3\eta}(\beta_{12})^{\frac{3}{2}}+\frac{\overline{N_2}}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta Q_1}\gamma_{1\overline{2}}-\frac{N_2}{3\eta}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta Q_1}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}+\frac{N_2\overline{Q_2}}{3\eta Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\right\}\left[\left.\frac{df_1(x)}{dx}\right|_{x_1}-\left.\frac{df_1(x)}{dx}\right|_{\overline{x_2}}\right]$$

$$\begin{aligned}&=\left\{\frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}+\frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}\gamma_{1\overline{2}}-\frac{N_2}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta {Q_1}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}+\frac{N_2\overline{Q_2}}{3\eta {Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\right\}\left.\frac{df_1(x)}{dx}\right|_{x_1}\\&-\left\{\frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}+\frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}-\frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}(\gamma_{1\overline{2}})^2-\frac{N_2}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta {Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right.\\&\left.+\frac{N_2\overline{Q_2}}{3\eta {Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^2\right\}\left.\frac{d\overline{f_2(x)}}{dx}\right|_{\overline{x_2}}\end{aligned} \tag{142}$$

Adding all three sections

$$\begin{aligned}&=\frac{4}{7\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(\gamma_{1\overline{2}})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}-\frac{4}{7Q_2}(\beta_{12})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2}+\frac{4}{7Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}-\frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}\\&-\frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1}+\frac{5N_2}{21\eta Q_2}(\beta_{1\overline{2}})^3\frac{df_2(x)}{dx}\Big|_{x_2}-\frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2(\gamma_{1\overline{2}})^3\frac{df_2(x)}{dx}\Big|_{x_2}+\frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\\&+\frac{5N_2Q_2}{21\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{5\overline{N_2Q_2}}{21\eta {Q_1}^2}(\gamma_{1\overline{2}})^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{5N_2Q_2}{21\eta {Q_1}^2}(\gamma_{1\overline{2}})^2\\&\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}+\frac{5N_2\overline{Q_2}}{21\eta {Q_1}^2}(\gamma_{1\overline{2}})^2(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}+\frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5\overline{N_2Q_2}}{21\eta {Q_1}^2}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5\overline{N_2}}{21\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}\\&+\frac{5N_2}{21\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5N_2}{21\eta Q_1}(\beta_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}-\frac{5N_2\overline{Q_2}}{21\eta {Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1}+\frac{5N_2Q_2}{21\eta {Q_1}^2}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\Big|_{x_1}\\&+\frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}-\frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_2(x)}{dx}\Big|_{x_2}\\&+\frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{df_2(x)}{dx}\Big|_{x_2}+\left\{\frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}+\frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}-\frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}\gamma_{1\overline{2}}-\frac{N_2}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}-\frac{N_2Q_2}{3\eta {Q_1}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}\right.\\&\left.+\frac{N_2\overline{Q_2}}{3\eta {Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\right\}\frac{df_1(x)}{dx}\Big|_{x_1}-\left\{\frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}+\frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}-\frac{\overline{N_2Q_2}}{3\eta {Q_1}^2}(\gamma_{1\overline{2}})^2-\frac{N_2}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}\right.\\&\left.-\frac{N_2Q_2}{3\eta {Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)+\frac{N_2\overline{Q_2}}{3\eta {Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^2\right\}\left.\frac{d\overline{f_2(x)}}{dx}\right|_{\overline{x_2}}\end{aligned}$$

$$= \frac{4}{7\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(\gamma_{1\overline{2}})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} - \frac{4}{7Q_2}(\beta_{1\overline{2}})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}}$$
$$- \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} + \frac{5N_2}{21\eta Q_2}(\beta_{1\overline{2}})^3\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2(\gamma_{1\overline{2}})^3\frac{df_2(x)}{dx}\Big|_{x_2} + \left\{\frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\right.$$
$$\left. - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right\}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$- \frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}(\gamma_{1\overline{2}})^2\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{5N_2Q_2}{21\eta{Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$+ \frac{5N_2\overline{Q_2}}{21\eta{Q_1}^2}(\gamma_{1\overline{2}})^2(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \left\{\frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}} - \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}\right.$$
$$\left. + \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}(\gamma_{1\overline{2}})^2 + \frac{N_2}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}} + \frac{N_2Q_2}{3\eta{Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{N_2\overline{Q_2}}{3\eta{Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^2\right\}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\Big|_{x_1}$$
$$+ \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5\overline{N_2}}{21\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5N_2}{21\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{5N_2}{21\eta Q_1}(\beta_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}$$
$$- \frac{5N_2\overline{Q_2}}{21\eta{Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{5N_2Q_2}{21\eta{Q_1}^2}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\Big|_{x_1} + \left\{\frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}} + \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\gamma_{1\overline{2}}\right.$$
$$\left. - \frac{N_2}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta{Q_1}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}} + \frac{N_2\overline{Q_2}}{3\eta{Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\right\}\frac{df_1(x)}{dx}\Big|_{x_1} \tag{143}$$

At second,

$$K_{11}\cdot 2 = \int_0^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_2(x'))^{\frac{3}{2}}}{\sqrt{x'}} \tag{144}$$

$$= \int_0^{x_2} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{x_2}^{\overline{x_2}} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}} + \int_{\overline{x_2}}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\int_x^{x_1} dx'\frac{(f_1(x'))^{\frac{3}{2}}}{\sqrt{x'}}$$

Substituting values from appendix

$$= (\beta_{12})^{\frac{3}{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\left[\frac{(\beta_{12})^{\frac{3}{2}}}{Q_2}\left\{\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{df_2(x)}{dx}\right\} + \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left\{\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_2(x)}{dx}\Big|_{x_2}\right\} + \frac{1}{Q_1}\left\{\frac{df_1(x)}{dx}\Big|_{x_1}\right.\right.$$
$$\left.\left. - \gamma_{1\overline{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right\}\right] + (\gamma_{1\overline{2}})^{\frac{3}{2}}\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\left[\frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\left\{\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{d\overline{f_2(x)}}{dx}\right\} + \frac{1}{Q_1}\left\{\frac{df_1(x)}{dx}\Big|_{x_1} - \gamma_{1\overline{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}\right\}\right]$$
$$+ \int_{\overline{x_2}}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\left[\frac{1}{Q_1}\left\{\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{df_1(x)}{dx}\right\}\right]$$

$$= \frac{(\beta_{12})^3}{Q_2}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} - \frac{(\beta_{12})^3}{Q_2}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx} + \frac{1}{\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$- \frac{1}{\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{Q_1}(\beta_{12})^{\frac{3}{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}$$
$$- \frac{1}{Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{\overline{Q_2}}(\gamma_{1\overline{2}})^3\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$- \frac{1}{\overline{Q_2}}(\gamma_{1\overline{2}})^3\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx} + \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{Q_1}\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{(\gamma_{1\overline{2}})^{\frac{5}{2}}}{Q_1}\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}}$$
$$+ \frac{1}{Q_1}\int_{\overline{x_2}}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{1}{Q_1}\int_{\overline{x_2}}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}$$

The second term, eighth term and last term gets cancelled with $K_{11}\cdot 1$,

$$
\begin{aligned}
&= \frac{(\beta_{12})^3}{Q_2}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&- \frac{1}{\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{Q_1}(\beta_{12})^{\frac{3}{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} \\
&- \frac{1}{Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}\int_0^{x_2} dx\sqrt{x}(f_2(x))^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{\overline{Q_2}}(\gamma_{1\overline{2}})^3\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&+ \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{Q_1}\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{(\gamma_{1\overline{2}})^{\frac{5}{2}}}{Q_1}\int_{x_2}^{\overline{x_2}} dx\sqrt{x}(\overline{f_2(x)})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{1}{Q_1}\int_{\overline{x_2}}^{x_1} dx\sqrt{x}(f_1(x))^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{(\beta_{12})^3}{Q_2}\frac{N_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{1}{\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{N_2}{3\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{1}{\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{N_2}{3\eta}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{(\beta_{12})^{\frac{3}{2}}}{Q_1}\frac{N_2}{3\eta}\frac{df_1(x)}{dx}\Big|_{x_1} \\
&- \frac{(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}}{Q_1}\frac{N_2}{3\eta}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}^2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}^2}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&+ \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}}{Q_1\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{Q_1\overline{Q_2}}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}\frac{df_1(x)}{dx}\Big|_{x_1} \\
&- \frac{(\gamma_{1\overline{2}})^{\frac{5}{2}}}{Q_1\overline{Q_2}}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{(\gamma_{1\overline{2}})^{\frac{5}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{Q_1\overline{Q_2}}\left[x\frac{df_2(x)}{dx} - f_2(x)\right]_{x_2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&+ \frac{1}{{Q_1}^2}\left[x\frac{df_1(x)}{dx} - f_1(x)\right]_{x_1}\frac{df_1(x)}{dx}_{x_1} - \frac{\gamma_{1\overline{2}}}{{Q_1}^2}\left[x\frac{d\overline{f_2(x)}}{dx} - \overline{f_2(x)}\right]_{\overline{x_2}}\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{N_2}{3\eta}\frac{(\beta_{12})^3}{Q_2}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta}\frac{(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{N_2}{3\eta}\frac{(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}\frac{df_2(x)}{dx}\Big|_{x_2} + \frac{N_2}{3\eta}\frac{(\beta_{12})^{\frac{3}{2}}}{Q_1}\frac{df_1(x)}{dx}\Big|_{x_1} \\
&- \frac{N_2}{3\eta}\frac{(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}}{Q_1}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{N_2Q_2}{3\eta}\frac{(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}^2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{N_2Q_2}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{N_2Q_2}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{5}{2}} \\
&\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{\overline{N_2Q_2}}{3\eta}\frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}^2}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{N_2\overline{Q_2}}{3\eta\overline{Q_2}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&+ \frac{\overline{N_2Q_2}}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{N_2Q_2}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2\overline{Q_2}}{3\eta Q_1\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} \\
&- \frac{\overline{N_2Q_2}}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{5}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} - \frac{N_2Q_2}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{5}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \frac{N_2\overline{Q_2}}{3\eta Q_1\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{5}{2}}\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} \\
&- \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{N_2Q_2}{3\eta{Q_1}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{N_2\overline{Q_2}}{3\eta{Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{N_1Q_1}{3\eta{Q_1}^2}\frac{df_1(x)}{dx}\Big|_{x_1} \\
&+ \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} - \frac{\overline{N_2}Q_1}{3\eta{Q_1}^2}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{N_2Q_1}{3\eta{Q_1}^2}\left\{(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}} - (\beta_{12})^{\frac{3}{2}}\right\}\frac{df_1(x)}{dx}\Big|_{x_1} \\
&- \frac{N_2\overline{Q_2}}{3\eta{Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\Big|_{x_1} + \frac{N_2Q_2}{3\eta{Q_1}^2}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{N_2}{3\eta}\frac{(\beta_{12})^3}{Q_2}\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \\
&- \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{N_2Q_2}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{5}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{\overline{N_2}}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3 \\
&\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2Q_2}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{5}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \\
&+ \frac{N_2}{3\eta Q_1}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{5}{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{N_2Q_2}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \frac{N_2Q_2}{3\eta Q_1\overline{Q_2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{N_2}{3\eta Q_1}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{N_2Q_2}{3\eta{Q_1}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \frac{N_2\overline{Q_2}}{3\eta{Q_1}^2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{N_1}{3\eta Q_1}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \frac{N_2}{3\eta Q_1}\left\{(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} - (\beta_{12})^{\frac{3}{2}}\right\}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{N_2\overline{Q_2}}{3\eta{Q_1}^2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{N_2Q_2}{3\eta{Q_1}^2}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&= \frac{N_2}{3\eta}\frac{(\beta_{12})^3}{Q_2}\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \\
&+ \frac{\overline{N_2}}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{N_2}{3\eta Q_1}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{5}{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \\
&+ \frac{N_1}{3\eta Q_1}\frac{df_1(x)}{dx}\bigg|_{x_1}
\end{aligned}
\tag{145}
$$

Therefore, $K_{11} = K_{11}\cdot 1 + K_{11}\cdot 2$

$$
\begin{aligned}
&= \frac{4}{7\overline{Q_2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(\gamma_{1\overline{2}})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\bigg|_{x_2} - \frac{4}{7Q_2}(\beta_{12})^3\sqrt{x}(f_2(x))^{\frac{5}{2}}\bigg|_{x_2} + \frac{4}{7Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\bigg|_{\overline{x_2}} - \frac{4}{7\overline{Q_2}}(\gamma_{1\overline{2}})^3\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\bigg|_{\overline{x_2}} \\
&- \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\bigg|_{x_1} + \frac{5N_2}{21\eta Q_2}(\beta_{12})^3\frac{df_2(x)}{dx}\bigg|_{x_2} - \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2(\gamma_{1\overline{2}})^3\frac{df_2(x)}{dx}\bigg|_{x_2} + \left\{\frac{N_2}{3\eta}\frac{(\beta_{12})^3}{Q_2} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\right. \\
&\left.\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) + \frac{N_2}{3\eta}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2\frac{Q_2}{\overline{Q_2}^2} - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right\}\frac{df_2(x)}{dx}\bigg|_{x_2} + \frac{5\overline{N_2}}{21\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{5N_2Q_2}{21\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3 \\
&\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{5N_2}{21\eta\overline{Q_2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}(\gamma_{1\overline{2}})^2\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} - \frac{5N_2Q_2}{21\eta{Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} \\
&+ \frac{5N_2}{21\eta}\frac{\overline{Q_2}}{{Q_1}^2}(\gamma_{1\overline{2}})^2(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \left\{\frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}} + \frac{\overline{N_2}}{3\eta\overline{Q_2}}(\gamma_{1\overline{2}})^3 - \frac{N_2}{3\eta\overline{Q_2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\right. \\
&- \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}} + \frac{N_2}{3\eta Q_1}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{5}{2}} + \frac{N_2}{3\eta\overline{Q_2}}(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}} - \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}} \\
&\left.+ \frac{\overline{N_2}}{3\eta}\frac{\overline{Q_2}}{{Q_1}^2}(\gamma_{1\overline{2}})^2 + \frac{N_2}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{5}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} + \frac{N_2}{3\eta}\frac{Q_2}{{Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{N_2\overline{Q_2}}{3\eta{Q_1}^2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}(\gamma_{1\overline{2}})^2\right\}\frac{d\overline{f_2(x)}}{dx}\bigg|_{\overline{x_2}} + \frac{5N_1}{21\eta Q_1}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&+ \frac{5\overline{N_2Q_2}}{21\eta{Q_1}^2}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{5\overline{N_2}}{21\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{5N_2}{21\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} - \frac{5N_2}{21\eta Q_1}(\beta_{12})^{\frac{3}{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} \\
&- \frac{5N_2\overline{Q_2}}{21\eta{Q_1}^2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\gamma_{1\overline{2}}\frac{df_1(x)}{dx}\bigg|_{x_1} + \frac{5N_2Q_2}{21\eta{Q_1}^2}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\frac{df_1(x)}{dx}\bigg|_{x_1} + \left\{\frac{N_1}{3\eta Q_1} + \frac{N_2}{3\eta Q_1}(\beta_{12})^{\frac{3}{2}} + \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}} - \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\gamma_{1\overline{2}}\right. \\
&\left.- \frac{N_2}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}} - \frac{N_2Q_2}{3\eta{Q_1}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}} + \frac{N_2\overline{Q_2}}{3\eta{Q_1}^2}(\beta_{\overline{2}\overline{2}})^{\frac{3}{2}}\gamma_{1\overline{2}}\right\}\frac{df_1(x)}{dx}\bigg|_{x_1}
\end{aligned}
$$

$$
\begin{aligned}
K_{11} = \frac{4}{7}\left[\frac{\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^{\frac{5}{2}}(\gamma_{1\overline{2}})^3}{\overline{Q_2}} - \frac{(\beta_{12})^3}{Q_2}\right]\sqrt{x}(f_2(x))^{\frac{5}{2}}\Big|_{x_2} + \frac{4}{7}\left[\frac{(\gamma_{1\overline{2}})^{\frac{5}{2}}}{Q_1} - \frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}}\right]\sqrt{x}(\overline{f_2(x)})^{\frac{5}{2}}\Big|_{\overline{x_2}} - \frac{4}{7Q_1}\sqrt{x}(f_1(x))^{\frac{5}{2}}\Big|_{x_1} \\
+ \left[\frac{N_2}{3\eta}\left\{\frac{(\beta_{12})^3}{Q_2} - \frac{(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}} + \frac{Q_2}{\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2 - \frac{(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)}{\overline{Q_2}}\right\} + \frac{5N_2}{21\eta}\left\{\frac{(\beta_{12})^3}{Q_2}\right. \right. \\
\left.\left. - \frac{Q_2}{\overline{Q_2}^2}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)^2(\gamma_{1\overline{2}})^3\right\}\right]\frac{df_2(x)}{dx}\Big|_{x_2} + \left[\frac{N_2}{3\eta}\left\{\frac{(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}}{Q_1} - \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3}{\overline{Q_2}} + \frac{(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{5}{2}}}{Q_1}\right. \right. \\
\left. + \frac{(\beta_{12})^{\frac{3}{2}}(\gamma_{1\overline{2}})^{\frac{3}{2}}}{\overline{Q_2}} - \frac{Q_2}{\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{(\beta_{12})^{\frac{3}{2}}\gamma_{1\overline{2}}}{Q_1} + \frac{(\gamma_{1\overline{2}})^{\frac{5}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}}}{Q_1} + \frac{Q_2}{{Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{\overline{Q_2}}{{Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^2\right\} \\
+ \frac{\overline{N_2}}{3\eta}\left\{\frac{(\gamma_{1\overline{2}})^3}{\overline{Q_2}} - \frac{2(\gamma_{1\overline{2}})^{\frac{5}{2}}}{Q_1} + \frac{\overline{Q_2}}{{Q_1}^2}(\gamma_{1\overline{2}})^2\right\} + \frac{5}{21\eta}\left\{\frac{\overline{N_2}}{\overline{Q_2}}(\gamma_{1\overline{2}})^3 + \frac{N_2Q_2}{\overline{Q_2}^2}(\gamma_{1\overline{2}})^3\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) - \frac{N_2}{\overline{Q_2}}(\beta_{\overline{2}2})^{\frac{3}{2}}(\gamma_{1\overline{2}})^3\right. \\
\left. - \frac{\overline{N_2Q_2}}{{Q_1}^2}(\gamma_{1\overline{2}})^2 - \frac{N_2Q_2}{{Q_1}^2}(\gamma_{1\overline{2}})^2\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right) + \frac{N_2\overline{Q_2}}{{Q_1}^2}(\gamma_{1\overline{2}})^2(\beta_{\overline{2}2})^{\frac{3}{2}}\right\}\left.\right]\frac{d\overline{f_2(x)}}{dx}\Big|_{\overline{x_2}} + \left[\frac{N_1}{3\eta Q_1} - \frac{\overline{N_2Q_2}}{3\eta{Q_1}^2}\gamma_{1\overline{2}} + \frac{\overline{N_2}}{3\eta Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}\right. \\
+ \frac{N_2}{3\eta Q_1}\left\{(\beta_{12})^{\frac{3}{2}} - (\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}} - \frac{Q_2}{Q_1}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\gamma_{1\overline{2}} + \frac{\overline{Q_2}}{Q_1}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}}\right\} + \frac{5}{21\eta}\left\{\frac{N_1}{Q_1} + \frac{\overline{N_2Q_2}}{{Q_1}^2}\gamma_{1\overline{2}} - \frac{\overline{N_2}}{Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}\right. \\
\left.\left. + \frac{N_2}{Q_1}(\gamma_{1\overline{2}})^{\frac{3}{2}}(\beta_{\overline{2}2})^{\frac{3}{2}} - \frac{N_2}{Q_1}(\beta_{12})^{\frac{3}{2}} - \frac{N_2\overline{Q_2}}{{Q_1}^2}(\beta_{\overline{2}2})^{\frac{3}{2}}\gamma_{1\overline{2}} + \frac{N_2Q_2}{{Q_1}^2}\gamma_{1\overline{2}}\left(\frac{\beta_{12}}{\gamma_{1\overline{2}}}\right)\right\}\right]\frac{df_1(x)}{dx}\Big|_{x_1}
\end{aligned}
\tag{146}
$$